

Degree Project in Industrial Design Engineering

Second cycle, 30 credits

Enhancing Ergonomics and Usability in Battery Energy Storage Solutions: A Service Design Approach

KARIN BODIN
EDKA CORONADO OMANA

Enhancing Ergonomics and Usability in Battery Energy Storage Solutions: A Service Design Approach

Karin Bodin
Edka Coronado Omana



Master of Science Thesis TRITA-ITM-EX 2024:225
KTH Industrial Engineering and Management
Machine Design
SE-100 44 STOCKHOLM

KTH Industriell teknik och management

Examensarbete TRITA-ITM-EX 2024:225

Förbättrad ergonomi och användbarhet i batterilager: ett tillvägagångssätt för tjänstedesign

Karin Bodin
Edka Coronado Omana

Godkänt	Examinator	Handledare
2024-06-27	Claes Tisell	Sara Ilstedt
	Uppdragsgivare	Kontaktperson
	Rebaba AB	Felix Kruse

Sammanfattning

Detta examensarbete eftersträvade att leverera ett designförslag för ett batterilager (BESS) inklusive rekommendationer för att stödja ergonomi och användbarhet. Projektet undersökte hur en BESS kunde utformas för bättre leverans, användarinteraktion och hållbarhetsaspekter av lösningen. Projektet genomfördes i tillsammans med Rebaba, ett nystartat företag som skapar BESS med hjälp av återanvända elbilsbatterier.

Double Diamond metoden implementerades i genomförandet av projektet. Projektet inleddes med en förstudie där litteratur inom ämnet utforskades. Förstudien genomfördes för att förstå elnätet, nuvarande BESS-lösningar, marknaden och ergonomin. Användarstudier genomfördes även i samband med 13 djupintervjuer och tre studiebesök. Insikterna från förstudien analyserades och klustrades, vilket gav upphov till fyra nyckelinsikter. Den första insikten var att kunder har svårt att hitta en lämplig BESS-lösning då deras kommersiella byggnad/industri har begränsat med utrymme. Den andra insikten var att kunderna värdesätter tjänsten som en BESS tillhandahåller mer än den fysiska produkten eftersom de vill ta del av dess fördelar men saknar kunskap om hur man driver och underhåller systemen. Den tredje insikten var att kunderna har svårt att lita på batterisäkerheten eftersom de saknar kunskap om ny säker teknik. Den sista insikten var att kunder behöver skräddarsydda lösningar eftersom de har varierande intentioner och förutsättningar.

Idégenerering gjordes utifrån insikterna och tre potentiella koncept utvecklades. Element från varje koncept kombinerades för att bilda ett slutgiltigt koncept bestående av en webbsida och webbportal. Rebaba-webbsida har utformats för att hjälpa kunderna att introducera till BESS som använder andra generationens Elbilsbatterier och initiera bekväm och pålitlig kontakt med Rebaba. En webbportal utformades för att assistera i hanteringen av BESS-projekt och system. Det slutliga konceptet har en del förbättringsområden som kan identifieras genom att utvärdera konceptet och fokusera på utformningen av specifika funktioner.

Nyckelord: Användarinteraktion, Projektering, Tjänstedesign, Digitalt Verktyg, Designförslag.

KTH Industrial Engineering and Management

Master of Science Thesis TRITA-ITM-EX 2024:225

Enhancing Ergonomics and Usability in Battery Energy Storage Solutions: A Service Design Approach

Karin Bodin

Edka Coronado Omana

Approved	Examiner	Supervisor
2024-06-27	Claes Tisell	Sara Ilstedt
	Commissioner	Contact person
	Rebaba AB	Felix Kruse

Abstract

This thesis project aimed to deliver a design proposal for a Battery Energy Storage System (BESS) including recommendations to support ergonomics and usability. The project investigated how a BESS could be designed for better delivery, user interaction and sustainability aspects of the solution. This project was conducted in collaboration with a start-up called Rebaba, who deliver BESS using reused electric vehicle batteries.

The double-diamond method was implemented to carry out this project. The project began with a research phase where literature on the subject was explored. The research was done to understand the electrical grid, current BESS solutions, the market, and ergonomics. User studies were also conducted through 13 in-depth interviews and three study visits. Interviews were held with project managers, potential customers, system engineers etc. and were approximately 45 min long. The study visits were at a developer, supplier and installation site. The insights from the research were analysed and clustered, giving rise to four key insights. The first insight was that customers struggle to find a suitable BESS solution because their commercial building/industry has limited space. The second insight was that customers value the service that a BESS provides more than the physical product because they want its benefits but lack knowledge about running and maintaining the systems. The third insight was, that customers find it difficult to trust the safety of batteries because they lack knowledge about new technologies regarding safety. The final one was that customers need tailored solutions because they have varying intentions and conditions which complicate the project.

Ideation was made based on the insights found in the research and three potential concepts were developed. Elements from each concept were combined to form a final concept consisting of a webpage and web portal. The Rebaba Website was designed to aid in introducing BESS using second-life EV batteries to customers and initiate comfortable and reliable contact with Rebaba. A web portal was designed to help manage BESS projects and systems. The final concept has areas that could be improved by evaluating the concept and focusing on the design of specific features.

Keywords: User Interaction, Project Planning, Service Design, Digital Tool, Design Proposal.

Acknowledgement

We are proud to present our master's thesis project during which we have had the invaluable opportunity to meet a collection of people with excellent knowledge of battery energy storage solutions. We would like to acknowledge everyone who has contributed to this project.

We thank the founders of Rebaba, Paula Runsten and Felix Kruse, for their guidance throughout the whole project. Their support has been vital for understanding the area of BESS and for contacting professionals on the subject.

We would also like to thank all the interviewees who participated in this project. The insights are largely credited to the interviews, study visits, and fostered discussions. Finally, we would like to acknowledge our supervisor at KTH, Sara Ilstedt for guidance and motivation throughout the project.

Karin Bodin and Edka Coronado Omana

Stockholm, June 2024

Table of contents

Introduction **1**

1.1 Background 2

1.2 Project Scope 2

1.3 Goal 2

1.4 Sustainability 3

1.5 Method 4

1.6 Delimitations 4

Discover **5**

2.1 Electrical Grid 6

2.2 Battery Energy Storage Systems 9

2.3 Market 22

2.4 Ergonomics 26

2.5 Interviews 32

2.6 Study visits 36

2.7 Website 42

Define **43**

3.1 Brand Analysis 44

3.2 Clustering 49

3.3 Customer Journey 49

3.4 Insights 50

3.5 Key Insights 55

3.6 Function Analysis 55

Develop 57

4.1 Design Thinking Method 58

4.2 Ideation Workshop 59

4.3 Circular Design Ideation 61

4.4 Concept Generation 63

4.5 Concept Selection 66

4.6 Concept Development 67

Deliver 68

5.1 Rebaba Website 69

5.2 Rebaba Portal 74

Discussion 83

6.1 Evaluation 84

6.2 Limitations 84

6.3 Recommendations for future work 85

6.4 Conclusion 85

8. References 87

Appendix A - Stakeholder Map i

Appendix B - Customer Journey ii

Appendix C - Functional Analysis iii

Chapter 1

Introduction

This chapter presents the background, scope, and goal, followed by how the project is contributing to sustainable development, and lastly the methodology and the delimitations of the project.

1.1 Background

The ethical aspects of battery production can be questioned both environmentally and socially. It incurs significant amounts of carbon dioxide, water, and, unfortunately, thousands of human lives yearly. Despite this, we discard electric car batteries that retain 70-80 per cent of their initial capacity when they are no longer sufficient for the car (Olsson Jeffery, 2023). Additionally, the renewable energy transition using eg. solar cells or wind turbines causes a variable supply of electricity in the grid due to the occurrence eg. of sunlight or wind.

Rebabas' vision is to create a simple, functional and industrialised solution for these batteries, connecting them to use as power banks for industries, commercial properties, and energy producers to aid in creating a more constant supply of energy. They plan to buy used lithium-ion batteries and modify them to store energy as BESS. The systems utilise rechargeable units that store energy and discharge it when needed (Siemens Energy, n.d.). BESS consist of one or more modules that allow the user to compose according to their power and capacity needs.

1.2 Project Scope

Investigate how a BESS can be designed for ease of installation and user interaction. The project should explore the usability and sustainability aspects of the solution. The focus will be on designing a solution that enhances ease of installation and user interaction, aligning with Rebaba's commitment. The findings will be presented as a design proposal including recommendations to support ergonomics and usability using human-centred design principles.

1.3 Goal

Deliver a design proposal for a BESS including recommendations to support ergonomics and usability.

1.4 Sustainability

There are a lot of benefits to a BESS that uses second-life batteries. The BESS and its functions can be directly connected to the UN sustainability goal 7: *Ensure access to affordable, reliable, sustainable and modern energy for all* (United Nations, n.d.). BESS could be an efficient solution to provide electricity in areas with unstable or no access to electricity. The UN reported that in 2021 675 million people lived in darkness with 4 in 5 of them living in Sub-saharan Africa. The BESS is also a way to ensure access to electricity in growing cities where the demand is high and only continuing to grow. Worldwide the solution would also make the shift to variable renewable energy easier.

The solution explored in this project also aligns with goal 9; *Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation*. Carbon dioxide emissions reached a rocketing 36.8 billion metric tons in 2022. At the same time, global manufacturing growth has slowed by 4.1% from 2021 to 2022 due to inflation, supply distributions, global economic deceleration and high energy prices (United Nations, n.d.). BESS is a way to support industries by reducing energy costs and making them more resilient to power outages. It is a way to ensure that production can continue and take advantage of more renewable energy.

Goal 12 is about *Ensuring sustainable consumption and production patterns*. The UN reports that high-income countries leave a larger environmental footprint than low-income countries. Material footprint by capita in high-income countries is more than 10 times that of low-income countries. Countries in Europe must take action to reduce their environmental impact. Rebaba uses second-life electric vehicle batteries, which take advantage of existing resources and minimise the need for new raw materials that might be mined in unethical ways. According to the UN, since 2016 company sustainability reports have tripled which shows the shift in society. Companies are being held accountable by consumers and regulations that set new standards to comply with if they want to stay relevant.

1.5 Method

The project will follow the double diamond process through stages; *Discover*, *Define*, *Develop*, and *Deliver*. It is initiated by discovering the problem through diverging research to get a broad understanding of challenges. Then, defining the problem by converging to insights and articulating the challenges to solve through design.

The second diamond is where solutions are created. Solutions are developed through diverging ideation methods and creating concepts based on the problem knowledge. Finally, a solution is designed and delivered as a final result (Design Council, n.d.). Figure 1. shows an illustration of the Double Diamond process.

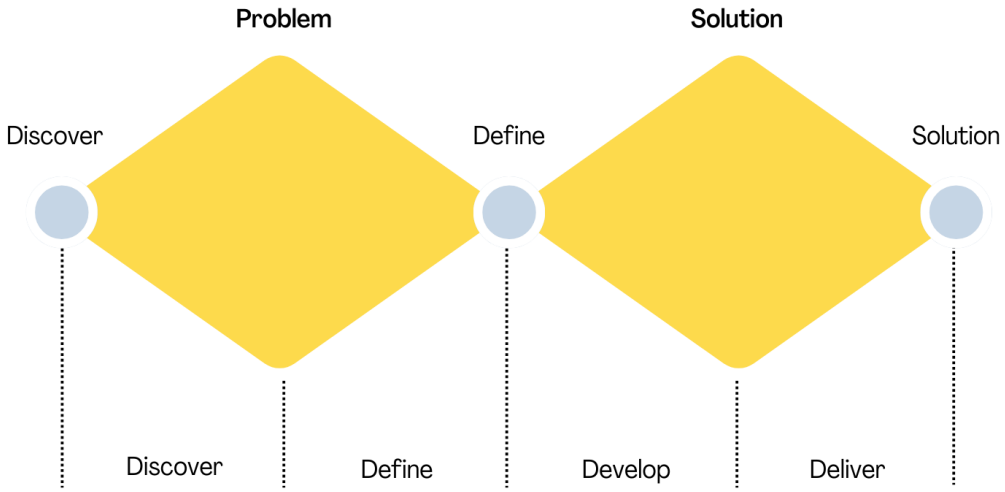


Figure 1. Shows the double-diamond method

1.6 Delimitations

This project will focus on the Swedish market and will explore battery energy storage systems for application in commercial and industrial (C&I) purposes. Systems in this sector typically range from 30 kWh to 10 MWh (McKinsey & Company, 2023).

Chapter 2

Discover

This chapter explores the problem area widely through an extensive literature study of various aspects connected to BESS.

2.1 Electrical Grid

The electrical grid in Sweden consists of transmission and distribution networks and connections for export. The electrical grid companies have exclusive rights to the grid making it impossible for other actors to choose which company to connect electrical systems to (Svenska kraftnät, 2022). However, the power grid consists of many parts and actors which can be separated into the physical parts for the electricity and the economic parts (Svenska kraftnät, 2021).

The electricity is transported from producers to consumers by networks. The transmission network runs through Sweden and transports large amounts of electricity from the energy producers to regional distribution networks. It utilises high electrical voltages, 400 or 220 kilovolts, to reduce power loss and distribute electricity in the distribution network through regional and local networks. Regional networks are connected to the transmission network and transport electricity directly to big energy consumers or the local network that connects to most electricity consumers like households and businesses. The electrical voltage on the regional grid is usually about 130 kilovolts and on the local grid less than 40 kilovolts. The voltage is lowered to 230 volts for the households connected to the grid. (Svenska kraftnät, 2022)

Electricity is traded in the economic part of the grid by monetary payments for the electricity used and transferred. Sweden is separated into four electric areas with different prices depending on their electricity market and the supply and demand. In the north of Sweden, there is an excess amount of electricity compared to the south where supply is low compared to the demand. Therefore, electricity is usually transported from the north to the south. (Svenska kraftnät, 2021)

2.1.1 Energy Sources

The Swedish power grid uses many types of energy sources providing a diversity necessary to regulate the balance between production and consumption. Sweden uses combined water, wind, sun, heat and nuclear power to supply the grid. The main sources of energy in Sweden are water and nuclear energy. These are plannable and regulated according to the energy need and provide stability to the grid. Figure 2 shows the distribution of energy sources in Swedish energy production. (Svenska krafnät, 2024)

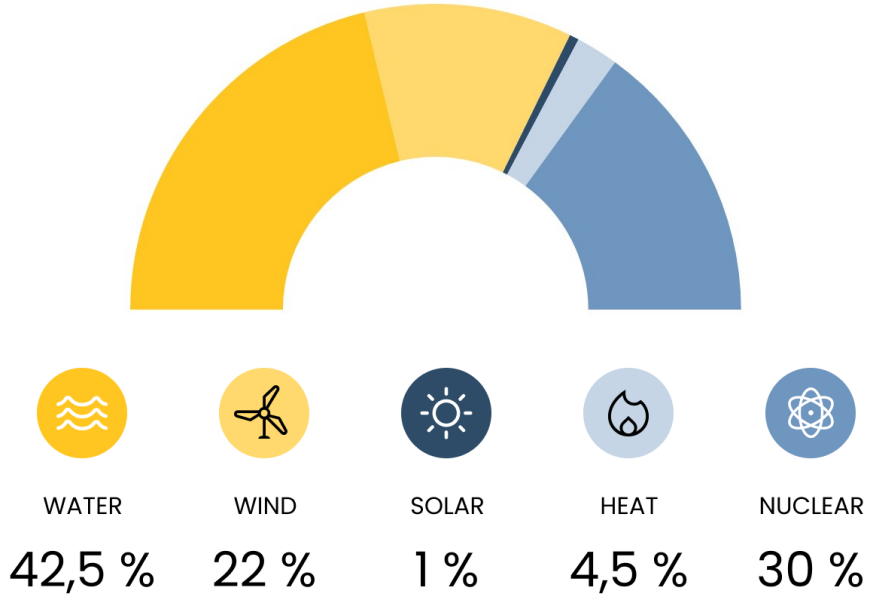


Figure 2. Energy sources in the Swedish energy production 2023. Not including household- and industry-produced electricity.

Lately, the country has been increasing the large-scale production of weather-dependent renewable sun and wind power. Since these sources are not plannable like water or nuclear power the grid needs balancing through voltage and frequency regulation. (Svenska krafnät, 2024)

2.1.2 Electrical grid imbalance

An electrical imbalance in the grid occurs when more energy is produced than consumed or vice versa. The frequency of the electrical grid signals the imbalance. If the frequency is below 50 Hz more energy is required, and if it is above 50 Hz less energy is required to balance the grid. The

energy needs to be produced the second it is necessary to balance the grid efficiently (Energiföretagen, 2023). Figure 3 demonstrates the principle of the energy grid balance between energy producers and consumers.



Figure 3. The required electrical grid balance between energy producers (to the left) and energy consumers (to the right).

2.1.3 Energy Storage Solutions

Energy storage is a way to create balance in the grid and postpone electricity consumption by storing energy until needed. When demand on the energy market is low the energy can be stored to be consumed when demand rises. Energy can be stored when it is cheap and used when it is expensive, or balance the energy grid when weather-dependent renewable energy cannot be produced. The size of an energy storage system is measured in megawatts which describes the amount of power that can be fed to the electric grid. The storage capacity is measured in megawatt hours, describing the total energy the storage system can hold. Energy can be stored in different forms eg. potential energy, kinetic energy, chemical energy or electric energy. In this project, batteries are used to store electric energy.

2.2 Battery Energy Storage Systems

A battery energy storage system (BESS) stores electricity from the grid in batteries. Depending on the intended use and placement, BESS vary in size, design and capacity. The system most commonly uses lithium-ion batteries because of their versatility (Balance Power, 2023). These are also common in electric vehicles and have lately been repurposed and applied in grid-scale solutions because of their remaining capacity.

2.2.1 Lithium-ion batteries

Lithium-ion batteries are at the centre of our modern world. It is popular due to its lightweight, high energy density and rechargeability, a key technology for limiting carbon dioxide emissions and slowing climate change (Deng, 2015).

A battery cell consists of a negative electrode (anode), a positive electrode (cathode), a separator, an electrolyte and two current collectors. In the battery the cathode stores the lithium while the electrolyte carries positively charged lithium ions from the anode to the cathode through the separator and vice versa. The movement of the lithium ions from the anode to the cathode releases free electrons in the anode creating a charge in the positive current collector. Discharging allows an electrical current to flow from the positive current collector through a device (such as an electric vehicle) and back to the negative current collector. The separator blocks the flow of electrons inside the battery. When the battery is charged the opposite happens.

2.2.1.1 Thermal Runaway

Li-ion batteries have a high energy density in terms of volumetric and gravimetric energy, which makes it possible for lithium-ion batteries to take many shapes while still being light compared to other batteries (Minos, 2023). However, there are some drawbacks regarding safety. Battery faults such as overcharging, electrolyte leakage, and external and internal short circuits could lead to thermal runaway. Thermal runaway is a non-reversible state where the battery experiences rapid self-heating. Upon the increased temperature, chemical reactions involving the battery materials generate gas and an increased pressure which could lead to cell rupture and release of toxic

flammable gases. Release and accumulation of these gases could result in an explosion. (Grönlund et al., 2023, 5). Figure 4 shows the thermal runaway of a battery.

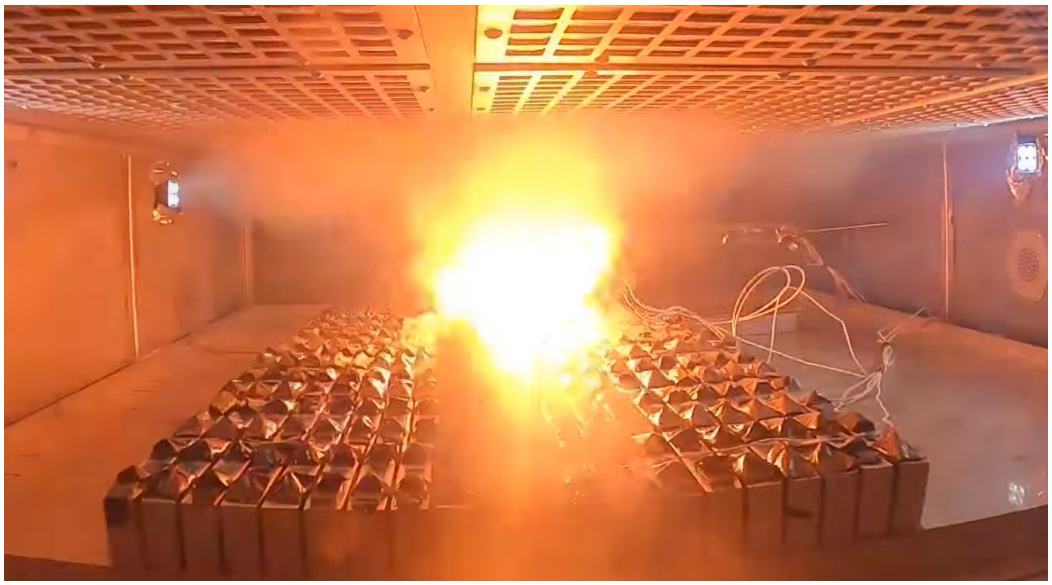


Figure 4. A screenshot from a test video showcasing thermal runaway (CellBlock FCS, 2022)

2.2.1.2 Lifetime

The performance of a battery can be evaluated using parameters such as cyclability. Cyclability measures the number of charge and discharge cycles before a battery loses energy significantly or can not sustain the function of the device it powers. The lifetime of a battery is particularly affected by the state of charge (SoC), depth of discharge (DoD), and operating temperature, in addition to the battery chemistry. SoC represents the percentage of charge contained in the battery, while DoD shows the percentage of discharge from the fully charged state. The more frequently a battery is charged and discharged the shorter its lifetime will be. Therefore, many battery manufacturers recommend a maximum DoD for optimal performance (Deng, 2015). The operating temperature for a Lithium-ion battery depends on the manufacturer and ranges between 10°C to 43°C (Senior, 2022). A too-hot environment makes the battery overheat and shortens its lifespan. A too-cold temperature makes the battery work extra hard at a higher voltage to charge (Deng, 2015).

2.2.1.3 Charge and Discharge Rate

The rate at which a battery is fully charged and discharged is described by its C-rate. Charging at 1C indicates that the battery is fully charged after 1 hour. Likewise, if a battery is fully discharged in one hour that would also indicate a C-rate of 1C. A higher C-rate indicates faster charging while a lower C-rate indicates slower charging. Figure 5 shows how the C-rates vary and their corresponding charge time.

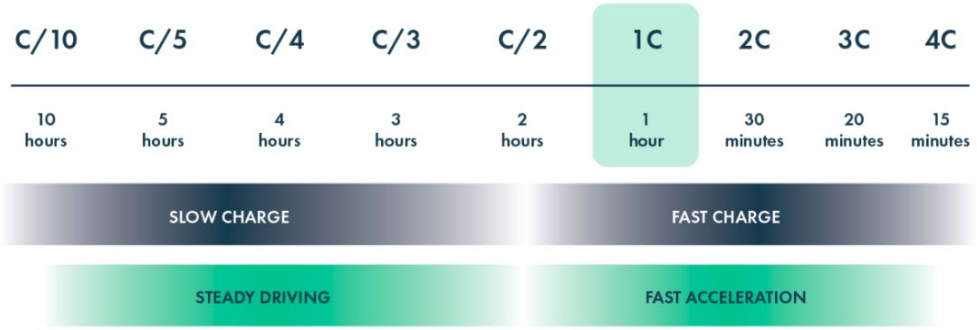


Figure 5. The scale of C-rate (Quantum Scape, 2021)

Higher C-rates increase the level of degradation in the battery which risks forming dangerous dendrites that cause cell failure and, in some cases, even fires. Therefore, chargers only sustain a charging rate higher than 2C for a couple of minutes before lowering the charging rate to avoid damage to the battery (Quantum Scape, 2021).

2.2.1.4 Material Extraction

The global strive for clean energy solutions has led to many advancements, however, as the demand for lithium-ion batteries increases it is important to acknowledge the environmental and ethical implications. Extraction of materials impacts the environment on site and globally. More than 60% of all lithium deposits are found in the *Lithium Triangle* between Chile, Argentina and Bolivia (Amnesty International, n.d.). The deposit areas are dry, making mining water-intensive, and leaving communities parched (GlobalData Thematic Intelligence, 2023). Extracting lithium also utilises sulfuric acid and sodium hydroxide which affect the soil and water and threaten to disturb ecosystems and endanger species. Indigenous people in the area have expressed concern surrounding the environmental impact of the mining as well as the working conditions (GlobalData

Thematic Intelligence, 2023). It has also been stated that carbon dioxide and greenhouse emissions from mining, extraction and production of lithium are more critical to the climate than the production of fossil fuel vehicles. Benchmark Mineral Intelligence estimates that the global demand for lithium, the key battery material, will quadruple and reach 3 million tonnes by 2030, exceeding supply. Most materials in an electrical vehicle battery are however likely to be recycled. Battery material waste over an electric car's lifetime will be about 30 kg by 2030, not including fossil fuel burned to generate electricity (Jolly, 2023).

In addition to the environmental impact of mining, children and artisanal miners are being exploited especially in parts of the Democratic Republic of the Congo at cobalt mines. Amnesty reported in 2016 that children 7 years old were digging for cobalt in narrow tunnels and carrying heavy loads. Workers have neither access to protective gear nor face masks, which causes them to inhale dangerous cobalt dust which could lead to serious health conditions. A later report from 2017, reviewed 29 of the world's leading companies and found that their supply chain for cobalt had improved. There are steps taken to ensure that the material used in batteries is being mined safely and humanly, but there is still a lot of improvement that could be done (Amnesty International, n.d.).

2.2.3 Electric Vehicle Batteries

An electric vehicle battery typically comprises thousands of rechargeable lithium-ion cells connected to form the battery pack. The three basic types of battery cells used in electric vehicles are cylindrical cells, prismatic cells, and pouch cells. Cylindrical cells similar to household AA or AAA batteries are the most commonly used format (EVBox, 2023). These are contained in a cylindrical casing that protects them against mechanical shocks. Prismatic cells are larger and can store more energy and deliver higher power than cylindrical cells due to the use of less material. Lastly, pouch cells, commonly used in cell phones, are encased in a soft plastic casing, making them space-efficient. However, this casing makes them fragile, needing additional protection to prevent mechanical damage. Figure 6 shows images of each cell type.



Figure 6. The three most common cell types; cylindrical cells, pouch cells and prismatic cells (Cliff Lee, 2023).

Another fundamental part of electric vehicle battery packs is their battery management system (BMS) which controls every aspect of the battery and ensures optimal performance. It monitors and regulates the charge level of each cell, decides which cells to charge or discharge, and keeps track of the temperature (EVBox, 2023). The batteries are also equipped with a cooling system to manage the heat produced by the cells. All components are contained in a protection case making the battery pack resistant to mechanical impact, waterproof and flame retardant. Figure 7 shows a battery pack for the Audi Q6 e-tron containing 15 prismatic cells.

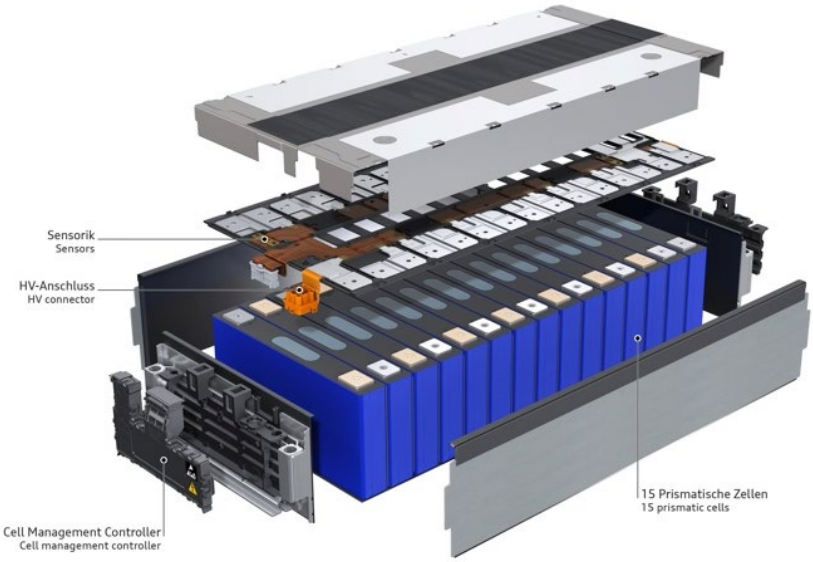


Figure 7. Audi Q6 e-tron Lithium-ion Battery Pack (EVKX, 2024).

2.2.4 Second-Life Electric Vehicle Batteries

When lithium-ion electric vehicle batteries reach the end of their life in vehicles, they often have between 70-80 per cent capacity left. Therefore, reusing them in other applications before reaching the end of their operational lifespan, and recycling them, can extract additional value. These batteries are referred to as Second-Life Electric Vehicle Batteries.

Adopting second-life batteries reduces landfill waste and the mining of new raw materials. In addition to the environmental benefits, it has proven to be more cost-effective than manufacturing new batteries. However, a challenge when using second-life batteries is adapting the BMS. A BMS for an electric vehicle is not suited for the intended second-life use, and every application demands a specifically engineered BMS to monitor and control the Lithium batteries, meaning that a battery module used in a BESS would need a new modified BMS. Another aspect that complicates the use of second-life batteries is the broad variation of battery modules in electric vehicles. Variations in form, chemistry and other factors may make mixing battery modules in a second-life application more difficult since it introduces unwanted variables (Guerra, 2023).

The lifetime of lithium-ion batteries ranges from 5 to 15 years and the cycle life from 1000 to 10,000. The amount of retired electric vehicle batteries is expected to increase due to the emerging green modes of transport. Chen and colleagues estimate that by 2030, 1 million electric vehicle batteries will be retired, and 1.9 million in 2040. Studies conclude that these are suitable for services requiring lower energy storage performance than vehicles, such as BESS. (Dong et al., 2023, 2).

2.2.5 EU Battery Directive

In July 2023 a new EU regulation was approved creating a unified legislation on the sustainability and safety of batteries in the EU (Swedish Chemicals Agency, 2024). The new regulation is part of *The European Green Deal*, Europe's growth strategy to transform the Union into a fair and prosperous society with a competitive economy and a net zero greenhouse emissions by 2050. Batteries are a key enabler of sustainable development, green mobility, clean energy and climate neutrality. Due to the strategic importance of batteries, the EU determined it would be necessary to set up regulations regarding the sustainability, performance, safety, collection, recycling and

second life of batteries etc. These regulations can be seen in the new 2023/1542 regulation. Some of the regulations apply directly while others are applied gradually.

Article 13 of the new derivative states that from 18 February 2027 all batteries with a capacity greater than 2kWh shall be marked with a QR code that contains information regarding the battery. The information should be updated and hold a battery passport with information regarding the capacity, state of health, prevention and management of waste. Batteries that have been re-used, repurposed or remanufactured shall bear new labels and information regarding their change of status should be accessible through the QR code.

Article 14 states that from 18 August 2024, up-to-date data regarding parameters determining the batteries' health and expected lifetime should be available through the BMS of stationary battery energy storage systems, Light Means of Transport (LMT) batteries and electric vehicle batteries. This is to evaluate the residual value, and lifetime and facilitate the re-use, repurposing and remanufacturing of the battery.

More details regarding the extended producer responsibility (EPR) are specified in the new Directive 2006/66/EC (Official Journal of the European Union, 2023). The EPR is a strategy to reduce the environmental impact of products and their packages. This will be done by making producers take responsibility for the whole product lifecycle (eg. take-back, recycling, disposal etc). In Sweden, The Environmental Protection Agency is assigned to enforce and review that regulations on producers' responsibilities are fulfilled nationwide (Naturvårdsverket, n.d.). Electric vehicle battery producers need to ensure the collection of waste batteries free of charge to the end-user. A producer established in another EU member state or a third country has to contract an authorised representative for EPR in each member state where the batteries are sold (Official Journal of the European Union, 2023).

Further, the derivative mentions requirements for testing for stationary battery energy storage systems using state-of-the-art testing methodologies to ensure the solution is safe. It is also written that removal and replaceability of LMT batteries, electric vehicle batteries and industrial batteries should be facilitated and easily done by independent professionals, which will take effect on 18 February 2027. The new legislation also requires manufacturers to affix the CE marking on their batteries before selling them or putting them into service, starting 18 August 2024. The CE

marking indicates that the battery complies with EU safety, health and environmental protection requirements (Sudhakar, 2023).

2.2.6 BESS Architecture

A BESS consists of a variety of components and sub-systems. Generally, it contains a battery rack holding battery modules and BMS, an inverter, and a controller. The system stands on a concrete base created to support the weight of the system and usually has an interface for the user to monitor the system. Figure 8 shows a simplified view of the BESS architecture.

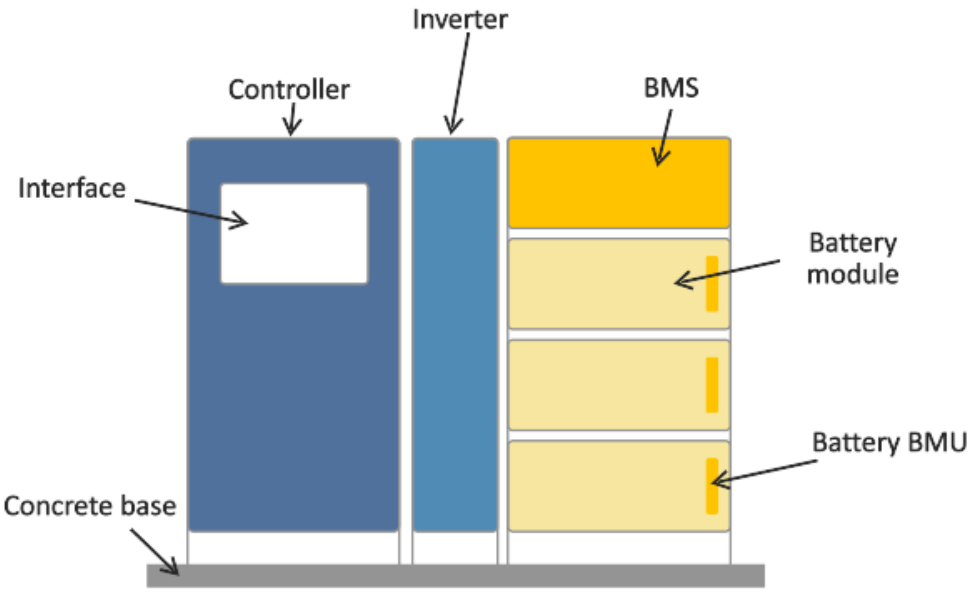


Figure 8.The basic architecture of a BESS.

2.2.6.1 Battery system

Repurposed electric vehicle battery packs are modified to be used in BESS. By replacing the individual Battery Management Unit (BMU) with a new one the battery pack is adapted to the new application. Battery packs are assembled in the system as modules. Rebaba calls their modules Baba Blocks and these charge and discharge DC electricity and weigh about 30 kg each. In a BESS these battery modules are connected in series and are mounted in a rack to form a large battery asset. The battery racks are then connected in series or parallel to reach the desired current and voltage of the BESS.

The BMS is the brain of the battery system and must be in any lithium-based storage system. The BMS is usually located at the top of the battery rack and safeguards and protects the batteries from damage. The BMS can achieve this by ensuring the battery operates within pre-determined ranges of critical parameters such as SoC, state of health, voltage, temperature and current. Most BMS allow for real-time monitoring and protection of the batteries at the cell and module level. The component uses application-specific algorithms to analyse data and control the battery's environment accordingly. A well-designed BMS is crucial for the safety and longevity of a BESS (Evesco, n.d.).

2.2.6.2 Inverter

The battery system is connected to an inverter to enable the charging and discharging of electricity from the grid. The battery system stores DC electricity and the grid provides AC electricity making conversion of current necessary. A power conversion system (PCS) or hybrid inverter is used to enable using battery energy storage. The PCS can be set to various modes of charging and discharging depending on the specific application of the BESS. For the inverter to know when to charge and discharge the batteries, it needs access to the status of the batteries. The PCS can be set by a pre-set strategy, such as on-site meters or an Energy Management System (EMS) (Evesco, n.d.).

2.2.6.3 Controller

The controller is the brain of the entire BESS, it monitors, controls, communicates and schedules. It communicates with external devices such as electricity meters and transformers ensuring the BESS runs optimally (Evesco, n.d.). The controller usually has an interface where users can see the system status.

2.2.6.4 Cooling

The system features a cooling system to keep an appropriate internal environment. Typically, through heating, ventilation, and air conditioning (HVAC) that cools by moving air between the inside and outside of the BESS using fans. Through thermal management, the life cycle of the battery system can be prolonged. Figure 9 shows what the HVAC can look like in a BESS.



Figure 9. Example of an HVAC cooling system in a BESS (Evesco, n.d.)

2.2.6.5 Fire distinguishing

A BESS is provided with a fire suppression system in case of thermal runaway of the batteries. Depending on the fire suppression technique, it could be activated by gas, smoke or temperatures. The fire suppression system will release an agent to suppress the fire and cool the system. It is usually designed depending on the size of the BESS.

2.2.7 Handling Risks

Risks associated with BESS depend on many factors including its environment, chemistry, and scale (e.g. power) and should be assessed accordingly. The design of the BESS and the planning around the project should accommodate the risks that may arise during its life cycle (IEC International Standard, 2020). The California Public Utilities Commission stated that approximately 2% of large-scale BESS in the United States encounter major safety events within two years of operating. 68 cases of BESS failure were reported in the energy storage technology database (EPRI) between 2011 and 2023 (Grönlund et al., 2023, 8).

2.2.7.1 Subsystems

During installation, it is important to ensure the function of critical subsystems for minimising risks such as the BMS, cooling and fire extinguisher. After the installation during commissioning, the subsystems shall be verified by proper testing to ensure their functions before the BESS is placed into service. Any subsystem that develops a fault which could affect the safe operation of the BESS should be isolated from other subsystems, however, the safety systems in place should not be affected by this isolation. The BESS design should focus on reducing noise, vibration and extreme temperatures, and make it easy for operators to identify hazardous parts, sections and conditions. The procedure to isolate a system should allow for secure handling of the different systems when disconnected, and the instructions to isolate the system should be easily accessible, provide clear instructions and checks to achieve isolation and provide clear instructions for restoring the system to operation. Figure 10 shows how the subsystems could be set up and isolated (IEC International Standard, 2020).

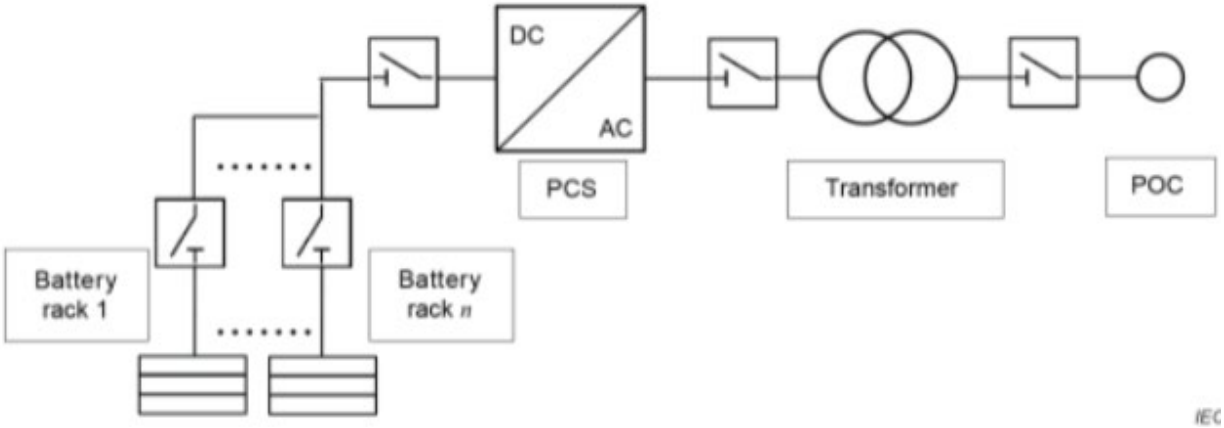


Figure 10. An example of how subsystems in a BESS can be isolated (IEC International Standard, 2020)

2.2.7.2 Fire Test Method

The greatest concern with BESS is thermal runaway in battery modules that could cause fires or explosions. The UL provides a methodology for conducting BESS fire tests called UL 9540A including a series of progressively larger fire tests at the cell level, module lever, unit level and installation level. The data from each test is used to evaluate the thermal runaway characteristics

and fire propagation. The complete data documentation is provided for authorities to determine the suitability of the BESS installation. Figure 11 shows the test sequence of the fourth UL 9540A edition. (UL Solutions, 2019)

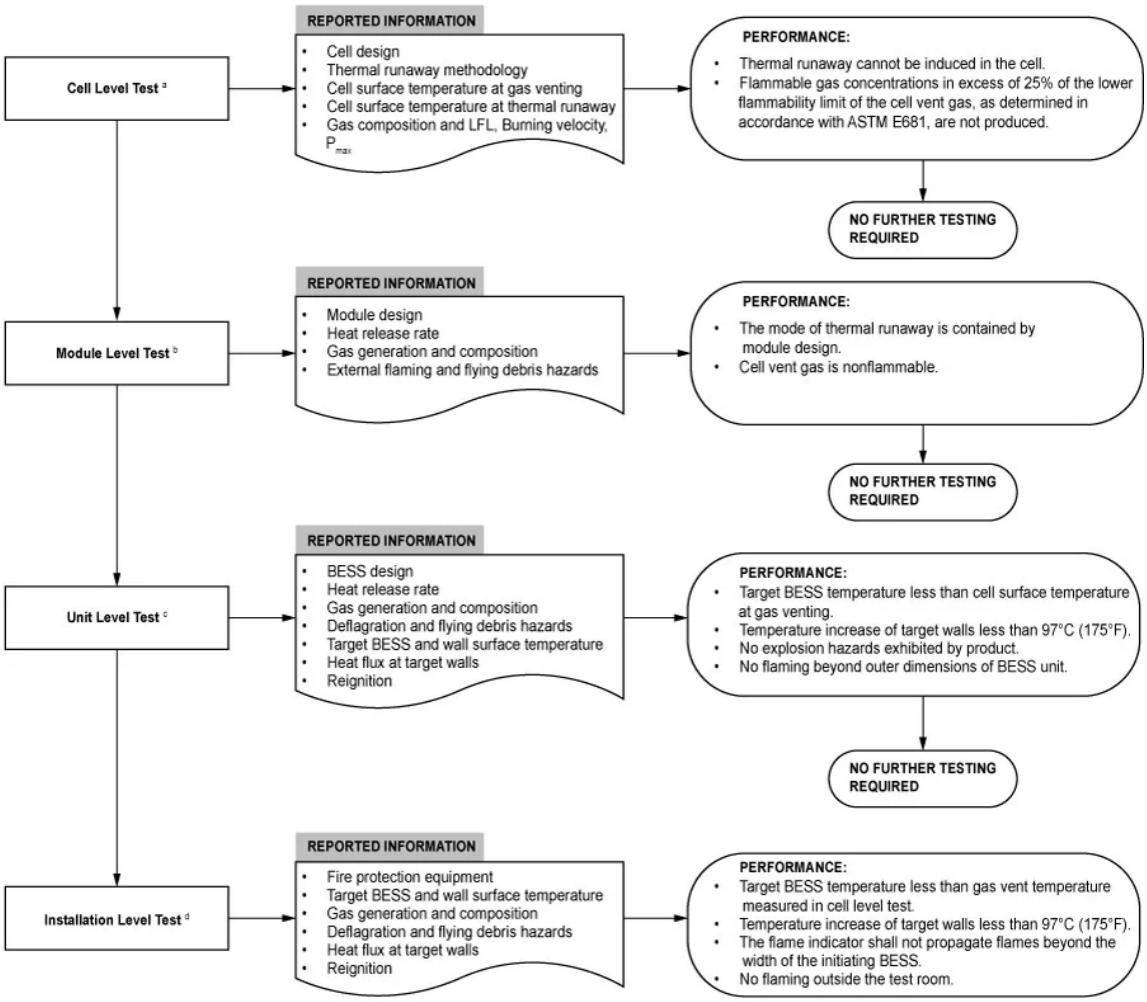


Figure 11. Test sequence of the UL 9540A test method (UL Solutions, 2019)

2.2.7.3 Safety Measures

An inherently safe design of BESS is the first and most important approach to reducing risk. Wherever the product design lacks safety, guards and other protective devices should be used. Additional equipment like an emergency stop mechanism might also be needed. Experience, however, shows that even well-designed guards and protective gear can be violated or fail. Therefore, combining approaches is beneficial. Another approach is information exchange

procedures where users comply with instructions given by the designer or supplier. The system should also warn the user of hazardous parts of the BESS using signs and signals, and safety information regarding the system should be available to all stakeholders.

A preventative maintenance schedule created by the manufacturer or system integrator is important to keep the system safe. It should consider the frequency of use, the time elapsed and the environment, and the maintenance should encompass the whole system. Reactive maintenance is also important for occurrences that impact the system integrity, such as water ingress due to unexpected weather conditions, or when animals impact the system (IEC International Standard, 2020).

2.2.7.4 Recommendations

There are no Swedish regulations, standards or guidelines regarding fire safety and installation of BESS. However, the Research Institutes of Sweden (RISE) has partnered with several stakeholders to develop a national guideline for BESS fire protection (Research Institutes Of Sweden, n.d.). Some recommendations for BESS in Multi-Dwelling Blocks or Businesses are as follows. They recommend that if the BESS is placed inside a building, it should be installed in a fire cell. If the system is placed outside, Bolagverket's Building Regulations (BBR) section 5:61 states an 8-meter separation between buildings should be applied, and alternative protection involves limiting fire spread through a fire cell or firewall standard in each building. The placement should also consider a safe distance to air vents and the risk of explosions associated with BESS. There are also recommendations for fire gas ventilation, which would enable the fire and rescue services to ventilate fire gases without opening doors that connect the BESS and the rest of the building. It could also be preferable to have an emergency response plan, containing information about placement, size, emergency break and fire safety as it could assist fire and rescue services (Grönlund et al., 2023, 34). Figure 12 shows an example of what a plan could look like.



Figure 12. Example of emergency response plan/ card (Grönlund et al., 2023, 34).

2.3 Market

The BESS market is relatively new and still growing. It is expected to reach USD 30.63 billion in 2024 and grow at a compound annual growth rate of 10.61% to reach USD 50.70 billion by 2029. Asia Pacific accounts for the largest market share and is estimated to grow at the highest rate over through 2024-2029 due to the growing energy demand in the area. The growth is supported by increased investments together with supportive government policies.

Factors such as declining prices of lithium-ion batteries and increased use of renewable energy may be driving forces in the market development of BESS. On the other hand, demand and supply mismatch of raw materials such as cobalt, lithium and copper may threaten BESS manufacturers that use first-generation batteries and rely on access to these materials to manufacture their BESS. The rapid technological development and research on new battery technologies may open new possibilities for the BESS market and lead to lucrative growth opportunities (Mordor Intelligence, n.d.).

2.3.1 Trends

The residential sector is expected to be the fastest-growing segment due to the rising investments in renewable energy infrastructure. Electricity consumption in residential buildings is estimated to increase along with the work-from-home trend and annual disposable incomes. Increased

technological advances in energy storage may lead to a decline in battery prices making renewable energy more accessible for residents. A huge demand for energy storage solutions comes from the solar energy sector as more residents are interested in renewable power sources. (Wu, 2023)

2.3.2 Competitors

Among the leading BESS producers are LG Energy Solution Ltd, Amp Nova, BYD and Tesla Inc, to name a few. A majority of competitors in the market supply BESS using first-generation batteries. Using second-life batteries is far more uncommon however some companies such as CATL share a similar product life cycle as Rebaba. CATL reuses some electric vehicle batteries in their BESS (Wu, 2023). The current market segmentation of BESS is highly competitive with many key players and the market rapidly growing. With many electric vehicle batteries expected to be retired in the coming years automotive companies such as Mercedes-Benz, Audi, Toyota and BAIC Motor are exploring the option of using second-life in BESS. The company B2U Storage Solutions operate a 25 MWh hybrid solar storage facility in Lancaster, California. The company uses a technology that enables the use of existing management systems in batteries, which reduces the repurposing costs significantly (Dong et al., 2023, 5).

2.3.3 State-of-the-art

CATL is a Chinese Li-ion battery manufacturing company specialising in developing electric vehicle batteries and BESS. Their revenue in 2022 was 56.7 Billion USD, a 22% increment from the previous year (GlobalData, n.d.), and they produced 43.4% of the world's energy storage batteries (Maisch, 2023). CATL is known as a system integrator and supplier to other system integrators who combine CATL's products in their BESS projects with their software solutions. This is due to political restrictions in countries like the United States, which prohibit using energy management systems (EMS) made in China (Murray, 2024).

CATL offer a variety of BESS solutions for different applications. In May 2022 CATL's EnerOne, an outdoor Liquid Cooling Battery System, won the EES Award for outstanding Innovations in the Storage Industry. The batteries in the BESS have a discharge time of 1C and a cycle life of up to 10,000 cycles. EnerOne can be shipped as a complete product which reduces installation time. Furthermore, it is suitable for outdoor applications due to the protection level of IP 66 on vital

components and an internal temperature range of -30 to +55 °C (CATL, 2022). An image of EnerOne can be seen in Figure 13.



Figure 13. CATL's BESS solution EnerOne (CATL, 2022).

CATL also offer containerised solutions for outdoor applications. EnerC is a 6058x2462x2896 mm containerised BESS weighing 35 tons. It has a modular design with a separate battery and electrical room for easy maintenance. Figure 14 shows the EnerC (CATL, n.d., 15).



Figure 14. CATL's containerised BESS solution EnerC (Ecactus Solar UK, n.d.).

In April 2024 the company launched *TENER*, a grid-scale BESS with 6.25 MWh per 20-foot container that assures zero degradation over the first five years.

2.3.4 Rebaba Solution

In the podcast *Lejonkulan*, the CEO of Rebaba Paula Runsten describe their innovative cooling system. It is at 50 times higher efficiency than market-standard HVAC. It is also self-extinguishing which is beneficial when using second-life batteries that get more hazardous with time. The technique allows for a longer system lifetime and fewer components which lowers the overall cost of the system. (Dahlqvist, 2024)

2.3.5 Stakeholders

A map of stakeholders was created with the customer in the centre. From the customer, various stakeholders such as Rebaba, assemblers, installers etc. were marked out. The value that is transferred between the stakeholders was illustrated using different icons. The stakeholder map can be seen in Table 1 in Appendix A. Identifying the stakeholders was crucial for determining suitable interviewees and understanding the relationship between different parties. It was also noted that there are many different stakeholders involved in delivering a BESS, contributing to the complexity of the product.

2.3.6 Use cases

The research has concluded in six typical use cases. Each use case has its own needs and the solution is tailored accordingly. Some parameters that would be adjusted are system size, capacity and C-rate. Table 1 lists these use cases ordered according to relevance for this project. Many customers will use BESS for a combination of the use cases.

Table 1. Use cases

1	Ancillary
2	Load shifting
3	Peak shaving
4	Power reserve
5	Use and store variable electricity

Load shifting is an electricity management technique that shifts load demand from peak to off-peak hours while keeping the total consumption constant. Peak hours are usually when machines and facilities are turned on in the morning and evening. This puts a lot of strain on the power grid, making it difficult to provide stable electricity for everyone. BESS can be used to support industries or facilities to ensure they have enough power to operate in these situations (gridX, 2024).

BESS is also used for peak shavings where the highest level of energy consumption during a short period is shaved. A peak can, for instance, occur when many electric vehicles are being charged simultaneously, which can strain the fuse and lead to a power outage. Peak shavings also reduce the company's expenses since they can maintain a lower energy consumption by using stored electricity from the BESS (gridX, 2024).

However, the most prominent use case in terms of value for the customer has been using the BESS for ancillary services. By using the BESS to partake in the support service market with frequency regulation customers can earn back their investment 10 times as much (Dahlqvist, 2024).

Customers who conduct socially important operations may also use BESS as a power reserve to maintain operations during a power outage. Finally, BESS are used for storing variable electricity from self-owned solar panels. Facilities that store variable electricity can power parts of their business or the entire business with green sustainable electricity.

2.4 Ergonomics

The project aims to create a solution for improved ergonomics during installation and use. Rose and Eklund (2019) present the definition, “Ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data and methods to design to optimise human well-being and overall system performance.”. Further, ergonomics is divided into three fields; *load*, *cognitive* and *organisation ergonomics*, each providing principles for improved

system performance. It is important to have a holistic approach when designing for the principles of ergonomics (Rose & Eklund, 2019, p.1-2).

2.4.1 Load Ergonomics

Load ergonomics treats the musculoskeletal system and injuries on it. Musculoskeletal disorders often occur in heavy lifting, repetitive work, and uncomfortable, strenuous work postures and movements. Working in sharply bent, stretched or twisted work postures puts a lot of strain on joints, which in those positions are more exposed. A high pace, stressful work and poor visual conditions can lead to poor work postures which can lead to musculoskeletal disorders. Acoustic conditions in the workplace also affect ergonomics. Workers risk putting too much pressure on their vocal cords if the acoustic conditions are unsuitable for speech. Vibrations can also cause disorders such as white finger syndrome, which is why it is important to use effective gear. Musculoskeletal disorders can be seen in all industries, both men and women are affected by them in manufacturing and assembly industries.

Static muscle work is when muscles work without a break or external movement. This type of work can be seen on the assembly line where the shoulders often work statically during repetitive work with the hands. During static muscle work, the blood flow to the muscles deteriorates, increasing the risk for musculoskeletal disorders. The body is built and thrives during movement but it is important to allow for recovery and engage in varied movements (Swedish Work Environment Authority, 2020).

2.4.3 Cognitive Ergonomics

The principle of cognitive ergonomics refers to how humans interpret, understand and process information. It is about adapting technical systems to human abilities and limitations. A good, customised solution can enhance the user experience, improve work performance and create a healthier and safer environment. The cognitive design principles described by Dan MacLeod (MacLead, 2013) can be used to increase the usability of a design and provide a more ergonomic solution.

The first principle when creating an ergonomic solution is to follow industry conventions and standards. People's interaction with the different products on the market set their expectations. Failing to maintain consistency may increase users' cognitive load by forcing them to do something new. To avoid confusion similar devices should work the same way. Figure 20 shows a screw-on cap for a bottle that follows the standard. Users know by experience that the cap opens when rotating the knob counterclockwise and closes when rotating clockwise.



Figure 20. The cap of a bottle follows the expectations of the user.

Stereotypes are another way to lessen the cognitive load of users. A stereotype is a common expectation among users of what is supposed to happen when they recognise a signal or activate a control. While standards are formally agreed upon to eliminate inconsistencies, stereotypes are less consciously determined. They develop over time and can differ depending on culture. Generally, red is associated with stop or danger and flipping switches upwards turns them on. Figure 21 exemplifies the lamp switch that follows a stereotype.

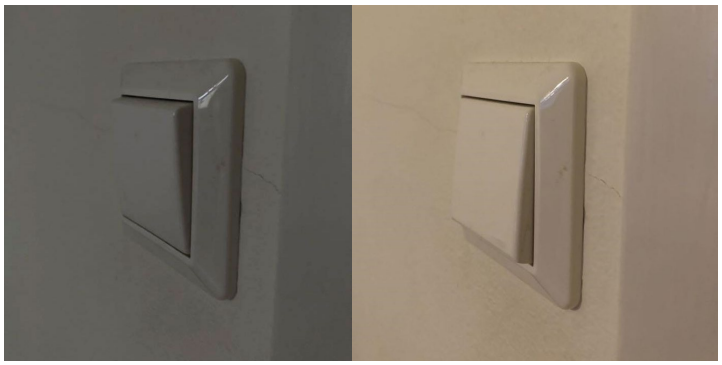


Figure 21. A lamp switch follows the stereotype.

Another way is to make sure the controls match the equipment layout. There should be compatibility between the perception of the need to take action and the action itself. In other words, there is a strong relationship between information display and control. When the controls correspond to the desired outcomes, it is easier for the users to learn how the interface works.

Providing too much information also increases the users' cognitive load. The design should have simplified displays of information that do not require careful examination and double-checking. The information should also be presented in appropriate detail meaning that the amount of detail should correspond to the users' need for information. The user should not be required to sort out what parts of the provided information are important. Another way to provide information to the user is by utilising clear images. The image should be clear and easy for the user to interpret. It should be visible in terms of placement, size and contrast to the background. It should be distinguishable from other surrounding information by utilising eg. sounds or visuals and be easily interpretable by the receiver.

The principle of redundancies is important to avoid mistakes. Sometimes, more than one message is needed for the user to assimilate information. It is important to provide the same information in more than one way. Figure 22 shows a stop sign placed in road intersections using three redundancies colour, shape and text.



Figure 22. Stop signs use three redundancies colour, shape and text.

Humans by nature are good at detecting patterns. Therefore, utilising patterns provides a quicker and more accurate understanding of the design. Control panels for complex equipment often utilise

patterns to provide information more clearly by grouping appropriate controls for ease of use. Controls can also be placed in their context to help users indicate their actions. Figure 23 shows the buttons of a TV remote control sorted after a pattern where apps are together.



Figure 23. The apps on this TV remote control are sorted after a pattern

Also, human senses fatigue easily with continuous input of stimulus making us less susceptible to information. Therefore, novel stimuli should be used eg. as warnings. Written warnings are often ignored since they become a part of the surroundings. Figure 24 shows an emergency vehicle that exploits this principle by switching sound and light alerting patterns to evoke attention more effectively.



Figure 24. A police vehicle switching to blue light and turning on sirens (Polisen, 2022).

Lastly, instantaneous feedback should be given to the user that tells about the success of an action. By providing feedback quickly the design makes it easy for the user to determine if an error has occurred and prevents it from being repeated. Figure 25 shows the provided feedback when placing the PlayStation controller onto its charging dock. The controller lights up when placed correctly and charging.



Figure 25. The feedback is given by the controller when placed in the dock.

Developing controllers for systems should generally encompass the principles of *Importance, Frequency,* and *Sequence*. If the principles contradict each other, it is the designer's role to decide which principle to prioritise (Rose & Eklund, 2019, p.24-26). Table 2 describes the three main principles for developing controllers, interfaces etc. named by Rose and Eklund (2019).

Table 2. Principles for controllers, interfaces and so forth.

Principle	Description
Importance	The most important controllers or functions should be most accessible
Frequency	The controller or function that is most frequently used in the work should be most accessible
Sequence	The controls or functions used in a particular sequence must be placed in this sequence and grouped

2.4.4 Organisational Ergonomics

Work organisation within ergonomics refers to how work tasks are divided between staff and how the tasks are structured. It also refers to how the goals of the organisation are formulated, how the work is directed, and how it is followed up and also includes social norms and cultures in the workplace (Rose & Eklund, 2019, p.28). Organisational conditions, such as time pressure, strongly controlled work and insufficient time for recovery, can contribute to musculoskeletal disorders. When workers experience a high mental load eg when a lot of information is processed or when decisions need to be quick muscle tension increases which can lead to discomfort.

2.5 Interviews

13 semi-structured in-depth interviews were conducted with stakeholders in the field to understand their experiences with BESS. The interviewees had varying roles such as project manager, chief technology officer, system engineer, and head of sales and provided insight into BESS projects and important aspects of their offering. The project format of BESS delivery to customers was described as a linear process including planning, installation, commissioning, and maintenance. Customers from C&I were interviewed to understand their needs and what values they see in a BESS solution. The BESS market is constantly changing and customers that invest in a BESS this early can be seen as risk-takers.

The interview findings are structured by the steps of the process of delivering value to the customer in Figure 26.



Figure 26. The value creation process for a BESS.

2.5.1 Planning

BESS projects are described as complex by the interviewees. Therefore, conducting rigorous planning with the customer and collaborative partners is important. Both project managers and sales professionals emphasise the importance of good documentation when communicating with parties.

There are many steps before delivering a BESS ready for startup. Before ordering a BESS the customer is in contact with the sales department to determine what solution best fits their use case, the intended placement for the BESS, and the power supply is assessed. An energy specialist at a property company in Stockholm expressed their need for systems adapted for their limited space. The company could not use BESS in its properties due to its constrained space and safety concerns.

After ordering a BESS different permits are applied for before preparing the site for installation, and the emergency services are notified about the project to evaluate the safety on site. FMEA (Failure modes and effects analysis) may also be used when planning the project to address potential risks and plan accordingly. The main challenge is having the right things on time which may delay the project and create customer dissatisfaction.

A BESS is heavy and needs a power supply from the grid to function. Therefore, the customer is asked to prepare the site according to standardised drawings provided by the supplier. A base should encompass the geological conditions in the site and is usually cast in concrete and leave holes for cabling.

2.5.2 Installation

The installation is purely mechanical and involves lifting, mounting, bolting, routing and connecting. The BESS is usually transported to the site by truck, lifted onto a prepared concrete base by crane, and bolted to the base to fixate it. The batteries are often transported separately from the cabinets and placed inside upon installation, using an electric lift. When the system is in place the cables are connected to get the system ready for commissioning.

The installation was described as quite fast and uncomplicated. However, unforeseen events such as disconnected or broken parts and weather can delay the installation. Some interviewees saw containerised solutions as an advantage as they are cost-effective and easy to transport ready

assembled. Others saw containerised BESS as a temporary solution that could be limited due to the set container size and better suited for customers like energy producers. Also, some integrators were working on standardising their solutions to make it easier for clients to understand their offering and also because it would make maintenance easier.

Electricians use manuals, drawings and documentation to support installation. The installing electricians mentioned that they learned how to install the system after interacting with a BESS from a specific supplier multiple times. Several also expressed that everyone interested can master how the systems work and should be installed.

The concrete base is sometimes constructed in a way that complicates installation. An interviewee described how the base can obscure the lifting of the batteries by hindering the electric lift from approaching the cabinet. Figure 27 shows a base that does not extrude far enough and creates a height difference in front of the cabinet obscuring the electric lift.



Figure 27. A concrete base that hinders the electric lift.

2.5.3 Commissioning

Commissioning is software-based and involves a lot of data handling, troubleshooting, and testing. The different parts of the system are put into service to confirm that they are working individually before connecting them and ensuring that the parts can communicate with each other. The system is connected to the grid when ready to confirm its intended function.

Interviewees noted that troubleshooting lacks set routines, but experience guides workers on what to watch for. A well-designed BESS uses flags and alarms to notify if something in the system is incorrectly connected or faulty and manuals are used to interpret them. Additionally, a clean user interface makes it easier to notice. Figure 28 shows how the error messages might appear in the control interface stating "*Charge_Error*" indicating possible problems with the inverter.

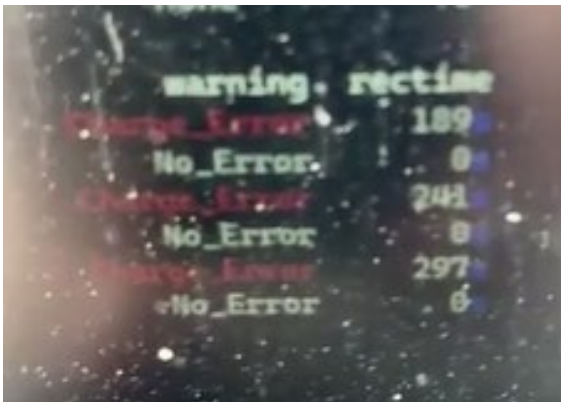


Figure 28. Error messages stating "Charge_Error" in red on the control interface.

2.5.4 Maintenance

Customers buy a BESS for its services instead of its physical presence. They expect the function of BESS to be maintained and is considered an important part of the company's offering. The system is to maintain its offering of services and therefore needs to be healthy and available. This involves monitoring the system and correcting issues if they arise. Most retailers offer a service agreement where they usually visit the customer to service the BESS once a year. During servicing, they among other things, check that the cooling system is working correctly, look for any noticeable damage on the battery modules, check for any water leakage, and do repairs. The value of servicing for the customer may vary, eg. electrical producers might not sign a service agreement since they know how to run and maintain the systems whereas C&I customers often need help with such services.

To determine the parts of a system that are important to be able to repair, integrators use MTF (mean time before failure) a metric used to evaluate which parts should be prioritised for repair and MSU (minimum swappable unit) to decide the level at which the system should be able to be

repaired. Redundant communication hardware is used to communicate to service personnel that something is faulty with the system.

2.6 Study visits

Three study visits were conducted to gain insight into delivering a BESS to the customer. During the study visits the interaction between the workers and the system was observed and challenges were identified.

2.6.1 Retailer

A retailer that sells and helps install the CATL systems was visited to explore their business and how they deliver the product to the customers. The responsible for their C&I offering introduced the CATL system and explained their experiences offering these systems to customers in Sweden.

The CATL systems are assembled in China and shipped ready for installation. The factory is informed about the Swedish Checkwatt requirements and is tested and adjusted accordingly. The system is delivered in big packages with the batteries separate from the cabinets. The system consists of battery cabinets, modules and a control cabinet featuring a control interface for the user. Figure 29 shows a system of one battery cabinet with five battery modules and a control cabinet.



Figure 29. A CATL BESS with one battery cabinet, five battery modules and a control cabinet.

When asked about installation, the employee explained that the power cables are challenging to handle. They are thick due to the high current and are difficult to guide into place when connecting. Figure 30 shows the power cable for the system they keep in their warehouse for testing.



Figure 30. The power cable to the BESS.

The system self-diagnoses and features a digital interface in the controlling cabinet where errors are displayed. Their control system is by CATLs daughter company Sunnic which caused a visual mismatch between the digital imaging and the real system. The Sunnic battery system is designed differently and when connected to a CATL system the interface still shows images of their system. Figure 31 shows the interface by Sunnic connected to the CATL system.

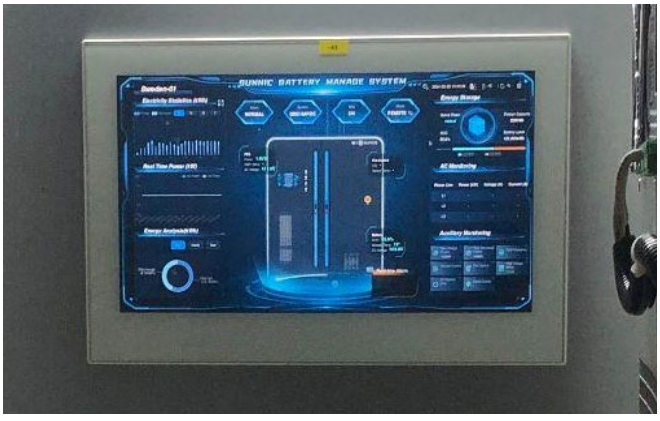


Figure 31. The Sunnic interface in the CATL control cabinet.

The home page of the interface displayed in Figure 31 above has a futuristic appearance and detailed information about the system's health was reached through a small icon in the top margin. When reaching the information interface the look of the screen changes into a more simplistic design displayed in Figure 32. The first page presents an overview of the basic health of the system. For more diagnostic information for the whole system, the user is met by a spreadsheet with lots of structured data.



Figure 32. The diagnostic interface of the control interface.

2.6.2 Supplier

A supplier of BESS using second-life electric vehicle batteries was visited to see how similar solutions are created. The company creates systems for electric boats using metal containers for the batteries and mounting them in racks. The battery containers are inserted into a rack made from extruded metal profiles and fastened using small screws.

The safety of using second-life electric vehicle batteries was described as a worry for some customers. They know of the safety hazards of battery accidents and want to ensure the system's stability. However, the company CEO explained that the batteries are treated with such care that they are not exposed to big risk factors. By controlling the batteries between 20-80%, refraining from high temperatures, and fast charging they are kept in good condition for a long time. He describes the BESS as a retirement home for second-life batteries.

2.6.3 Installation

An installation of a 500 kW BESS for an E-commerce pharmacy was visited to understand the work and limitations on site. The system for the facility consisted of 10 cabinets: one control cabinet with inverters, one bottom fixture for each, and about 126 batteries á 60 kg. The batteries were transported in separate cardboard boxes resulting in much waste and cleanup after installation. Figure 33 shows how the cabinets and batteries arrived packages on site.



Figure 33. Shows how the cabinets and batteries arrived packaged on-site.

Before installing the battery cabinets and control cabinets the cables are pulled up through the concrete base and the fixtures for the cabinets are mounted to the base. The installers explained that the concrete base was created incorrectly. Measurements differed from the standard drawings to which they had to adapt placing the fixtures. They explained that such mistakes are common and can be combated with planning and clear communication. Figure 34 shows the prepared concrete base with 10 fixtures for the cabinets mounted in place using expander bolts and conduits for the cabling.



Figure 34. Fixtures and cabling prepared for the installation of cabinets.

For this installation, a misunderstanding between the customer and the company caused a delay. The customer had built a roof over the site making it impossible for the installers to lift the cabinets onto the fixtures using a crane. Figure 35 shows the roof covering the site. The installers considered whether they could place the cabinets using a forklift or if the roof should be removed before proceeding with the installation.



Figure 35. The roof covering the site.

The installers discussed other possible hurdles when installing BESS. A factor that can affect the installation is the weather. Since the cabinets are normally lifted at the top using a crane, wind can cause the system to swing uncontrollably. Also, the cabinets cannot be connected when raining because the water will pool in the fixtures which need to be drained before connecting the cables.

During the study visit a plug-and-play solution was discussed to ease installation. The installers believe cables could be measured and cut beforehand and that a multi-connector would make the cable work easier. Having cables ready before installation would simplify their work and reduce waste. The power cable was also mentioned as hard to handle because of its thickness and weight. This system utilises five of these cables. They further explained that the cables are sensitive to bending which risks damaging the isolation. Figure 36 shows a snippet of the power cable used for the system.



Figure 36. The power cable for the system.

To further emphasise the size the installers extracted one conductor from the BESS power cable and one from a regular household power cable. Figure 37 shows the size comparison of the different conductors.



Figure 37. Comparison of the BESS power cable (top), household power conductor (middle), and BESS power conductor (bottom) size.

2.7 Website

Using dark mode in web design reduces devices' energy consumption and carbon footprint making the design more sustainable. Changing the colours according to their energy use can lower the screens' energy consumption using OLED (Organic Light Emitting Diodes) and LED (Light Emitting Diodes). Unlike LCD (Liquid Crystal Display) screens that light up all pixels using a backlight, OLED and LED screens deactivate individual pixels to display dark colours (Singh, n.d.). Using fewer pixels reduces energy use. Google compared the modes of Google Maps and found that dark mode reduces display power usage by 63% compared to normal (Tung, 2018). A research study at Purdue University, also found that switching to dark mode reduces the power draw by an average of 67% (Dash & Hu, 2021). Figure 38 shows a comparison of normal and dark modes in interfaces.

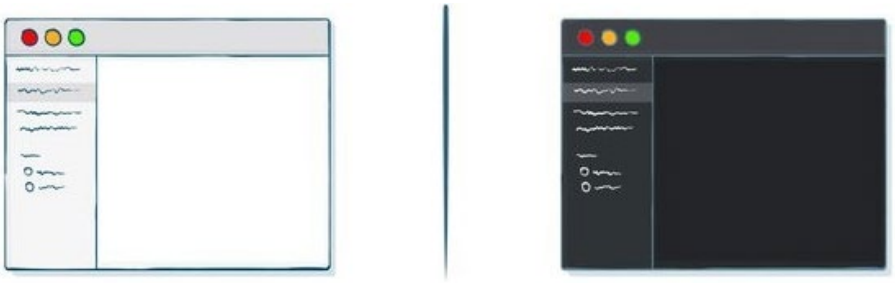


Figure 38. Normal mode (to the left) and dark mode (to the right) (Mahajan, 2017).

Chapter 3

Define

In this chapter, the information from the research, interviews and study visits were processed and formulated into insights. Clustering and customer journey mapping were used to sort through interesting observations, creating the base for six well-developed insights. Four key insights were created from the insight cards that present the most important insights.

3.1 Brand Analysis

The Rebaba brand identity was analysed to understand the brand better and to understand the aspects of the brand that would influence the BESS. Since the brand is relatively new it was also clear that choices regarding brand name, logotype, font and visual profile needed to be defined and contemplated to ensure the brand communicates intentionally. The analysis was based on the current state of the brand, and discussions with the co-founders. Figure 15 shows the brand analysis describing its *vision*, *mission*, *personality*, *values*, *offer*, and *benefits*.



Figure 15. Rebaba Brand Analysis

3.1.1 Brand Workshop

To explore the Rebaba brand a creative workshop was conducted with the co-founders. It allowed them to describe their brand in unconventional ways. Figure 16 shows pictures from the workshop together with Rebaba.

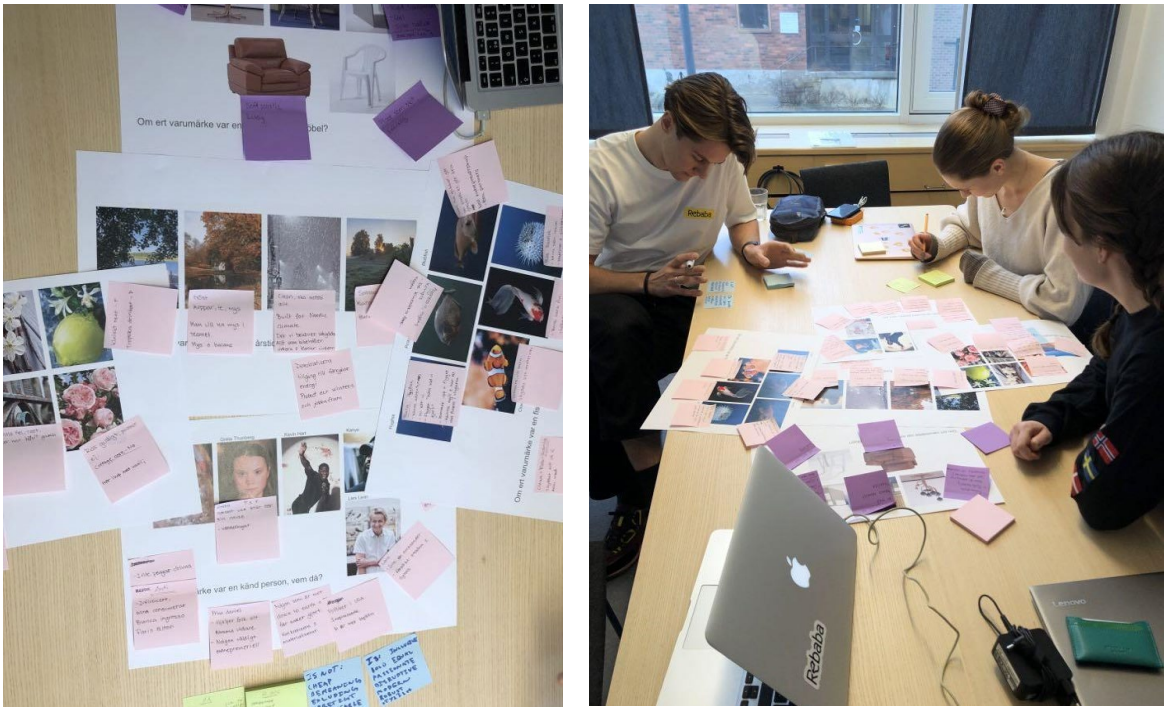


Figure 16. Pictures taken from the brand workshop.

The workshop presented collages of different categories to the participants of which they were encouraged to choose which described the brand best. Figure 17 shows one of the collages used in the workshop when picking a representative fish for the brand.

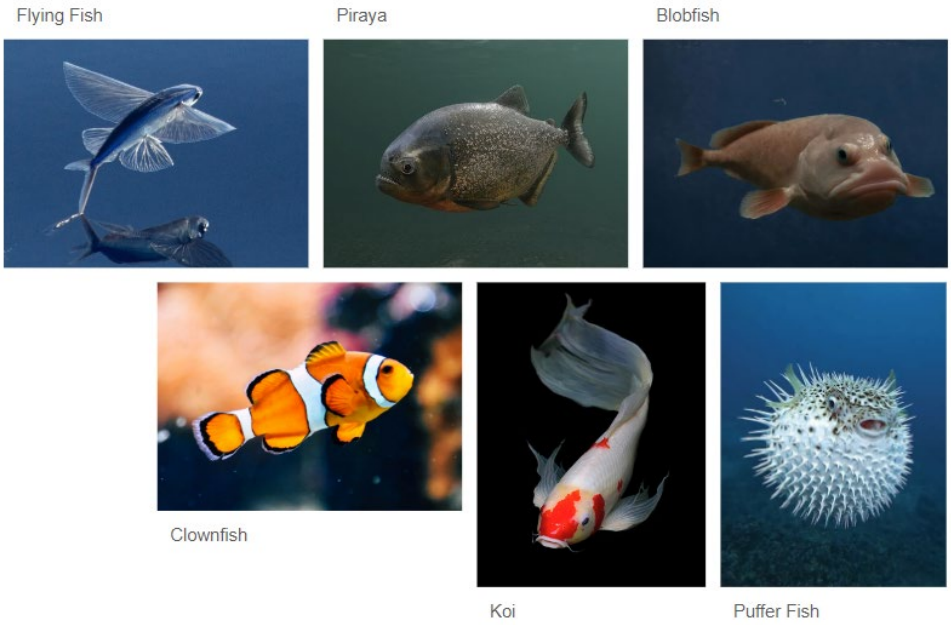


Figure 17. Example of images shown to the participants.

Each collage generated fruitful discussions about the brand and its values both in terms of its offering and its personality. The participants discussed the expression of each picture and explained what pictures suit the brand most and in what ways. The associations were then formulated into words that the founders believed to describe the brand. Some words were *transparent*, *genuine*, *emerging*, *agile*, *collaborative*, *clean*, *curious*, *diverse*, and *disruptive*. One of the founders concluded, "*We are not the last generation's stuff, we are what is to come*".

3.1.2 Brand in Words

Based on the discussions with the co-founders of Rebaba and the brand workshop the brand was defined using words that do and do not describe the brand identity. These are what the brand *is* and what the brand *is not*. Figure 18 shows the words describing the Rebaba brand.

IS	IS NOT
Reliable	Unprincipled
Visionary	Excluding
Benevolent	Cemented

Figure 18. The brand in terms of what it is or is not.

Rebaba is described as *Reliable*, *Visionary*, and *Benevolent*. The brand is reliable in its presence and offers long-term sustainable solutions of good quality. It is visionary and wants to push the limits of the market. They dare to challenge conventional ways in the belief of their values while being benevolent and honest, able to remain true to their goal and goodwill for society.

On the contrary, Rebaba is described as not *Unprincipled*, *Excluding*, and *Cemented*. Unprincipled means the brand is principle-driven and does not compromise its values for monetary gain. It is inclusive to all actors that want to help the cause and is therefore not excluding. They are curious to explore new ways and will not be cemented to traditional practices.

3.1.3 Brand Expression

Based on the knowledge about the brand a mood board was created to communicate the brand expression. The words describing the brand *Reliable*, *Visionary*, and *Benevolent* were visualised in a collage of pictures. It features an image showing two hands reaching towards each other against a partly cloudy sky capturing a moment that conveys a sense of yearning or reaching out for support or connection—further, an image showing a modern, uniquely designed chair made of metal. The chair has an angular, geometric structure with a minimalist aesthetic. Third, the board features an image showing a whale swimming underwater. It captures the majestic and serene

presence of the whale in its natural habitat. The Humpback whales are often perceived as curious and gentle giants. Mountains symbolise stability and endurance, as they have stood the test of time and weathered countless storms. Its high peaks reaching towards the sky evoke a sense of vision and aspiration symbolising lofty goals and forward-thinking. The mood board was created to support later ideation. Figure 19 shows the brand mood board for Rebaba.



Figure 19. Brand Moodboard.

3.2 Clustering

The interview and study visit findings were clustered using sticky notes to identify patterns and commonalities. The notes were clustered as shown in Figure 39 into topics like regulations, planning for a BESS, the cognitive aspects of the work and maintenance of the system to name a few.

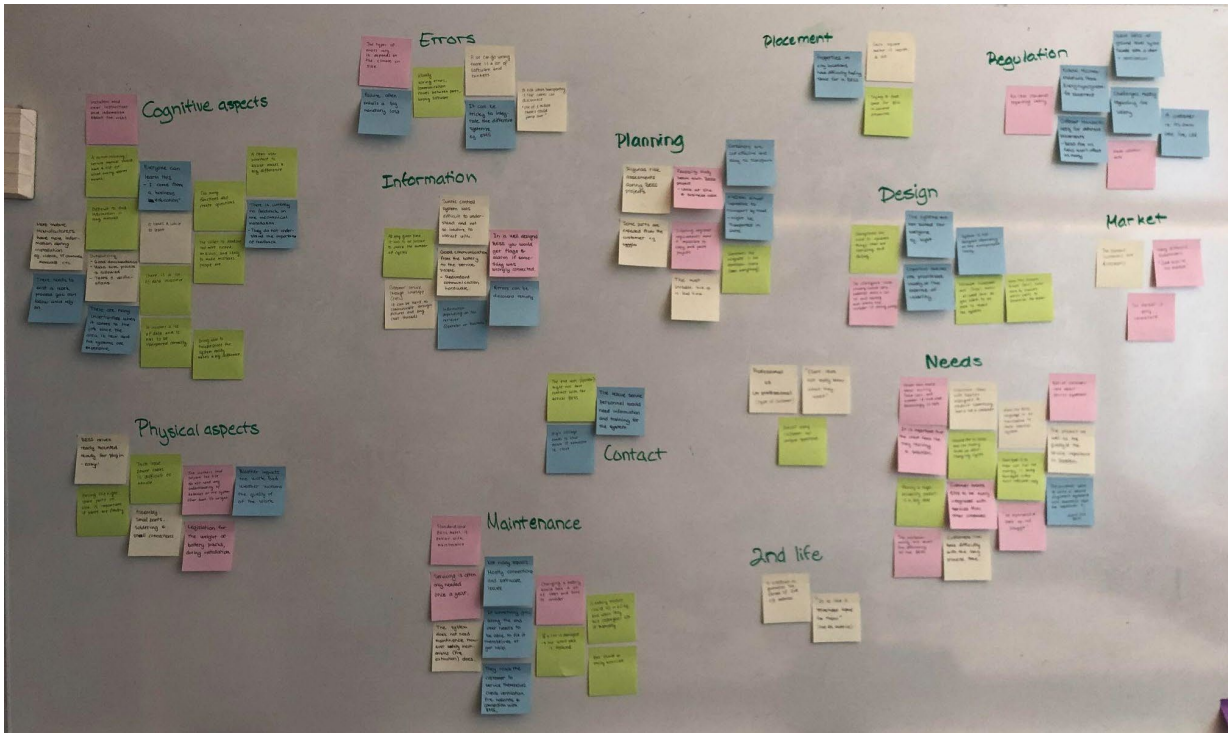


Figure 39. Clustering of interview and study visit findings.

3.3 Customer Journey

A customer journey of a BESS project was conducted based on the findings to recognise the challenges experienced by the customer. The different actions that the customer takes part in were lined up from planning to delivery. Needs and potential pain points during these actions were identified and the customer's feelings were mapped. It was noticed that negative emotions often occur when information is unclear such as when the customer is instructed to build a concrete base

or when they are thought to maintain the system. The customer involvement in the projects varies but they all share the need for clear communication and instructions. Confusion can also arise during the project since there are no clear standards or regulations due to the market being very young. The customer experiences great stress and anger if something in the system fails since it can result in off time and loss of quality. In this situation, the customer has to feel reassured and get quick assistance. The feelings throughout the BESS project can be seen in Figure 40. The complete customer journey can be found in Appendix B.

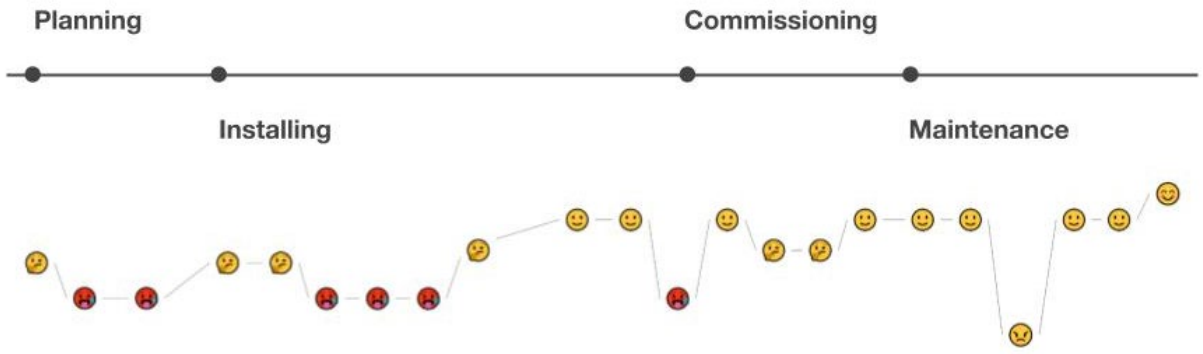


Figure 40. The customer's feelings through the delivery of a BESS.

3.4 Insights

After clustering and breaking down the findings from the research, it became evident that it was important to offer the product function to the customer and not the product itself. The customer group studied in this report are interested in the benefit of using a BESS but lacks knowledge and interest in the physical system. This realisation made it suitable to take a more service design approach to the insights and future work.

3.4.1 Insight Cards

To define the major findings retrieved from the interviews and study visits insight cards were constructed. The insight cards present each insight with a quote from one of the interviewees, a

statement defining the insight and a short description. Six insight cards were created and each will be presented in more detail.

The first insight card is seen in Figure 41. The insight is regarding the limited space that C&I have available to place a BESS. Industries and commercial facilities in the city might not own the ground around the facility or can be restricted due to densely built buildings. The option of putting the BESS indoors was discussed during interviews. However, buildings generally do not have space over, every square meter is accounted for and has an intended use. Interviewees expressed that they needed adaptable solutions that fit into small spaces, which have been difficult to find. Suppliers on the market today offer too big systems that might not be suitable to have indoors. When putting a BESS inside safety concerns become even more relevant and customers need a solution that can guarantee safety.

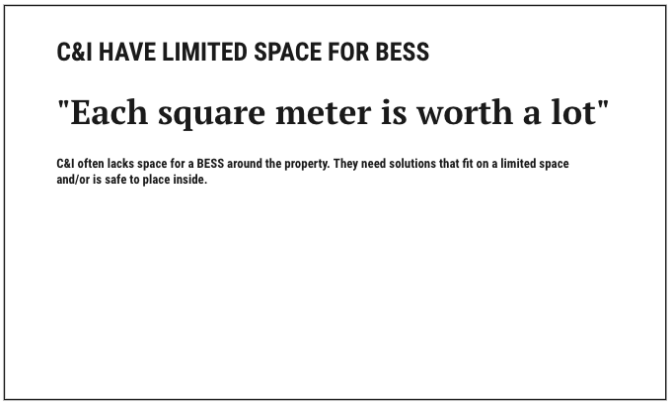


Figure 41. Insight card 1

The second insight card in Figure 42 states the possibility of simplifying the installation. With BESS installations and projects generally, there are a lot of parts that need to fall in place. The systems consist of various cooperating parts and for the installation to go smoothly workers need a reliable work process that minimises uncertainty. A product manager explained that uncertainties are due to the new market and lacking guidelines and standards. Concern also comes from the systems being expensive and workers avoiding mistakes that could be costly. Developing a plug-and-play solution would make the installation much smoother with wiring and connections.

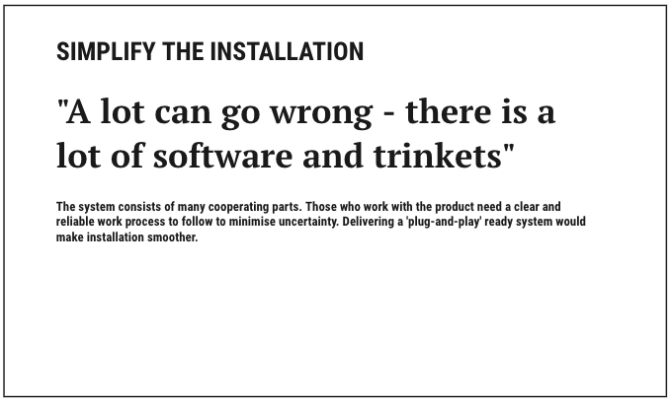


Figure 42. Insight card 2

The next insight seen in Figure 43 talks about the immature BESS market, where a system engineer said "*You reinvent the wheel often - each month*" referring to the fact that new developments happen rapidly. As mentioned earlier, there are no clear guidelines or standards for BESS and each company finds their way of working with the product and delivering it to customers. The current customers were described as risk takers by a product manager at a supplying company of BESS.



Figure 43. Insight card 3

The next insight is that each delivered BESS to a customer is a tailored project. This was realised early in the thesis project after interviewing various project managers at different suppliers. The insight can be seen in Figure 44. Varying regional regulations that might come from the rescue services and conditions such as space or connection to the grid require tailored solutions. The intended use of the BESS also impacts the delivery of the product and how it should be designed. These varying conditions and needs make it difficult to copy and paste projects from one customer

to another.



Figure 44. Insight card 4

Insight card 5, seen in Figure 45, concerns customers' uncertainty regarding lithium batteries and even more about second-life lithium batteries. People have heard of exploding electric vehicles and might have seen a video of a battery undergoing thermal runaway, which might be concerning when it comes to the safety of the batteries. There is a sense of distrust of the technology and customers as well as some suppliers believe that repurposed batteries are of lower quality. People need more information regarding safe technologies and understand how a second-life battery compares to a first-life battery. An energy specialist at a real estate company expressed he would like to know how a reused battery holds up after charging cycles and the estimated lifetime. Customers want to be reassured that it is worth the investment.

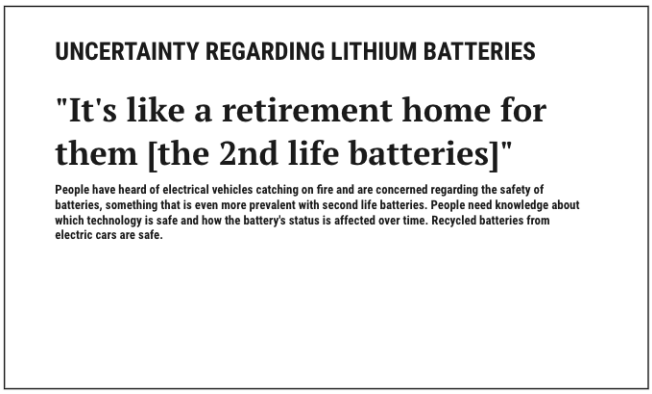


Figure 45. Insight card 5

The final insight card seen in Figure 46 states that people in Sweden expect service. A supplier of BESS said, “In Sweden the product as well as the quality of the service is important”. This is something that might not be true for every country or product but when it comes to this market in Sweden, customers want to feel confident that the BESS maintains its functions. The quoted supplier also mentioned that it is important to offer the same quality of service no matter the place in Sweden, which can be challenging in the countryside for example. A BESS should be seen as a product service system, delivering a well-designed service could make a BESS supplier stand out and create a closer and secure relationship with customers.



Figure 46. Insight card 6

3.4.2 Ergonomics

The relevancy of each ergonomics group was explored early in the project, after some research, it was discovered that work concerning BESS had more problem areas surrounding cognitive and organisational ergonomics rather than load ergonomics. Research was done on all three groups to maintain a holistic view of the area. However, it was decided that the most impact could be achieved by focusing on cognitive and organisational ergonomics, which became the main focus later in the project.

3.5 Key Insights

The insight cards were further defined formulating four key insights using the template. Table 3 lists the four key insights gathered from the research.

Table 3. The key insights from the research.

1	Customers struggle to find a suitable BESS solution because their commercial building/industry has limited space.
2	Customers value the service that a BESS provides more than the physical product because they want to get its benefits but lack knowledge about how to run and maintain the systems.
3	Customers find it difficult to trust the safety of batteries because they lack knowledge about new technologies regarding safety.
4	Customers need tailored solutions because they have varying intentions and conditions but it complicates the project.

3.6 Function Analysis

A function analysis was conducted to express the product service system through its functions. The analysis enables well-thought solutions in the developing stage of the process. The functions are described in a verb and a noun to isolate each function. Boiling them down to the most important opens for new radical solutions. The full functional analysis for the BESS is shown in Tabke 1 of Appendix C.

The main function of a BESS is to store electricity. Some of the subordinate functions were to provide transforming and conversion of electricity. For the main function, it is also important that the solution provides charging and discharging of the batteries, as well as managing (monitoring and controlling). The remaining functions were categorised as either *needs* or *wishes* depending

on their importance for the product's overall function. The functions were also grouped under subcategories such as ergonomics, target group and market. The functional analysis was deliberately formulated unspecific without clear parameters since it would be used to ideate from.

Chapter 4

Develop

This chapter following chapter presents how the insights from the previous chapter were used to form ideas to better the usability and ergonomics of BESS. Ideas were created using the How Might We design thinking method in a creative workshop and using the Circular Design pack of the Use2Use toolkit. The ideas were then compiled into three concepts which were developed into a final concept that was evaluated using usability evaluation methods.

4.1 Design Thinking Method

The How Might We design thinking method (HMW) was used to create innovative ideation sessions and help solve design challenges. The key insights were reframed and written as HMW questions to focus on more specific challenges. Table 4 shows the six HMW questions used when ideating.

Table 4. How Might We questions used in ideating sessions.

1	HMW create a BESS that fits commercial buildings/industries' limited space?
2	HMW encourage users to feel safe placing the BESS inside?
3	HMW minimise the off-time of the BESS?
4	HMW make customers feel safe using second-generation batteries?
5	HMW adapt the product offering to customers' varying intentions?
6	HMW adapt the product offering to customers' varying conditions?

4.2 Ideation Workshop

An ideation workshop was held with the founders of Rebaba to brainstorm ideas based on the HMW design thinking method presented in section *4.1 Design Thinking Method*. The workshop began by introducing the participants, doing a warm-up activity and the key insights before brainstorming ideas for each one of the HMW questions. Figure 47 shows the ideation workshop with Rebaba.

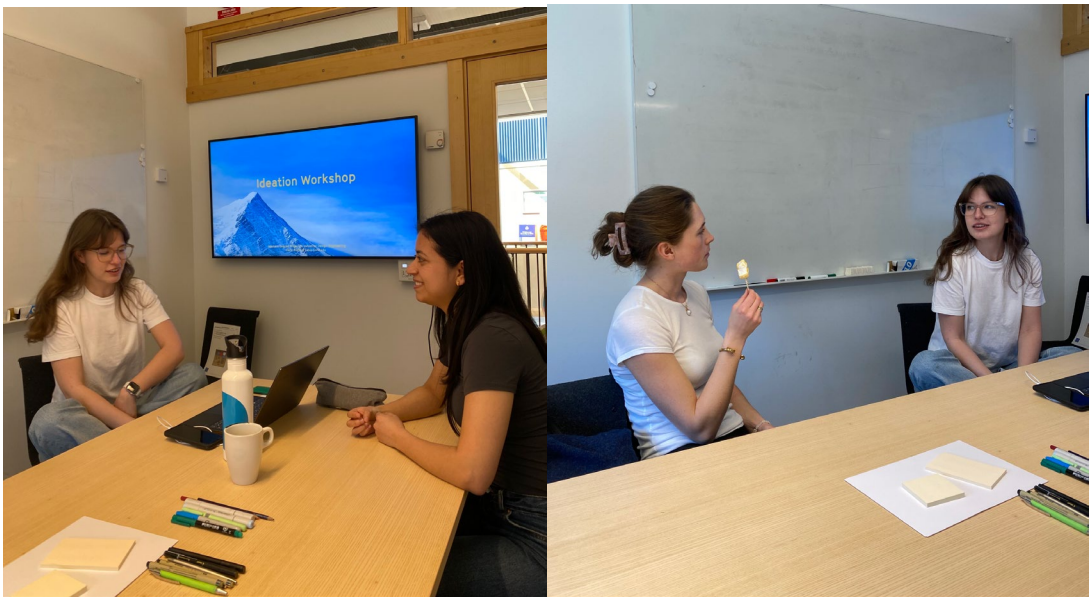


Figure 47. Ideation workshop with Rebaba.

The warm-up aimed to shift the focus from earlier activities to the workshop and to begin the ideation session in a positive and more creative state of mind. Each participant shared a good thing from their morning creating happy conversations. Then, the key insights in section 3.*4 Key Insight* were introduced to ensure all participants had a shared vision for the workshop outcome before starting the brainstorming.

HMW questions were presented one at a time to ideate in sessions. A timer was set for 10 minutes as everyone sketched or noted down ideas individually on a paper. After sketching everyone presented and discussed their ideas with the participant. The ideas for the six HMW questions were discussed and clustered by themes to clarify the solutions. Figure 48 shows the clustering by

themes *Maintain Value, Be a Buddy, Second Life Batteries, Communication of the BESS, Feeling Safe,* and *Fit Limited Spaces*.

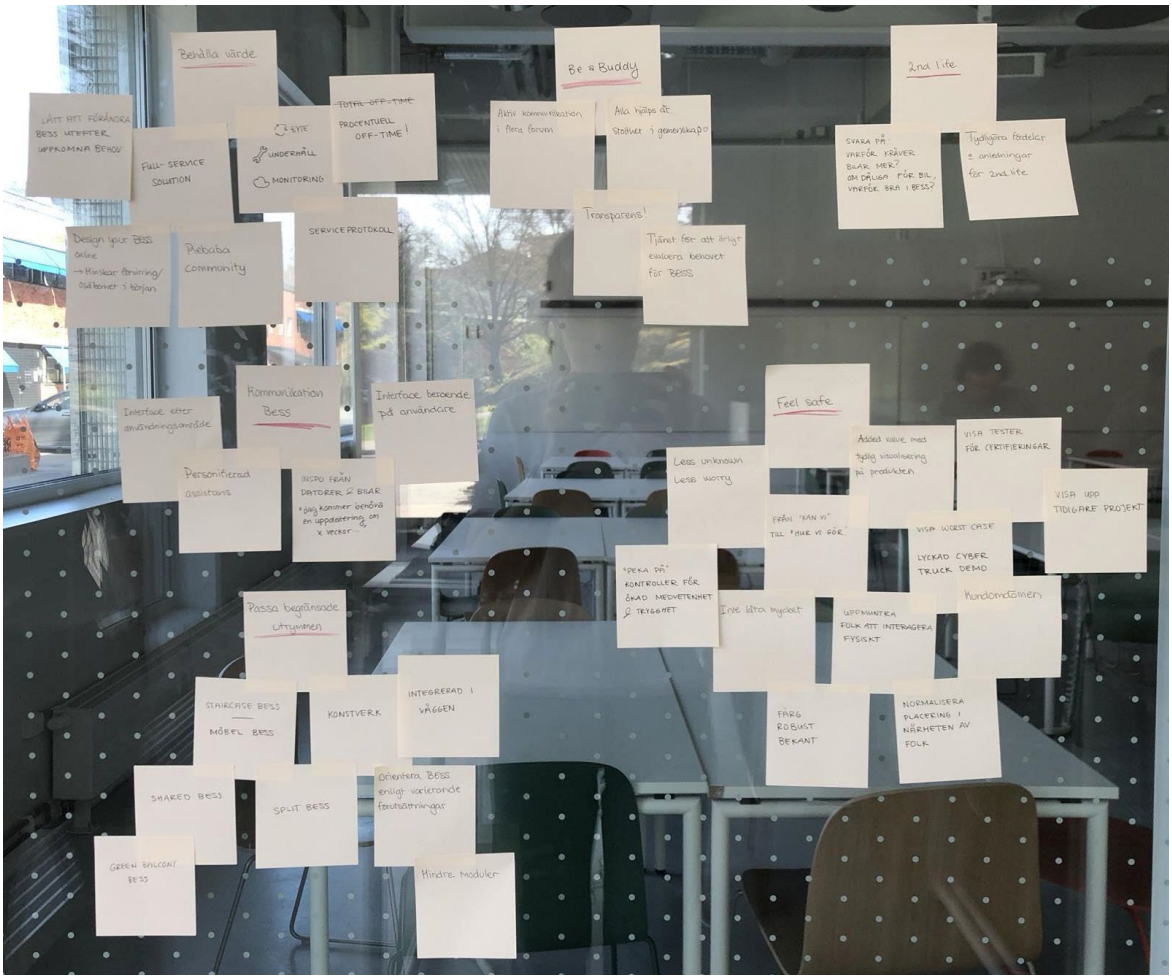


Figure 48. Clustering of the ideas generated in the workshop.

4.3 Circular Design Ideation

Introducing circular consumption solutions entails different practices and challenges compared to linear ones. Addressing these through design is essential to creating well-suited products and services for circular consumption. The circular designs ideation pack of the Use2Use Design Toolkit highlight design opportunities for creating a more circular design of the solution. It contains 38 strategy cards categorised into topics *Extended Use, Multiple Use-Cycles, Exchange,* and *Match-Making* which provide trigger questions to support ideation activities. The strategies were discussed to identify ways to address the design challenges and the main ideas were noted.

The Design for Extended Use strategy addresses the challenges associated with a product's reduced utility over time. A product's potential to fulfil users' needs may decrease over time due to technological shifts or new user needs. To encourage the use of the BESS for an extended period it should various ideas were discussed.

The use of the product can be extended by catering to different aesthetic preferences. From the research, it was concluded that C&I value the possibility of having the system placed indoors. One way of ensuring that the BESS fits in most environments is by giving it a minimalistic appearance. The BESS should also be compatible with most electric vehicle batteries to ensure the product stays relevant following technological shifts. To smoothly integrate the BESS with eg. solar panels from another supplier, the system should also be compatible with other software to optimise the use of the system. Frequent serving and maintenance can also keep the BESS in good condition, and to make it effective it is important to make it easy to identify potential errors with light or sounds for instance. Creating an experience where customers feel responsible for the BESS would also make them take better care of the system. Meaningful emotional bonds can also be formed for the product and service by making every interaction with Rebaba feel reliable and smooth, the customer should feel that they are taking part in a sustainable and meaningful solution and thus use the product for a longer time.

For a product to be circular it can also be designed for multiple use-cycles. The BESS that Rebaba offers reuses electric vehicle batteries which comes with some challenges, especially regarding customers' concerns with the quality and safety of the system. It could be beneficial to show customers how the safety system works to make customers more comfortable with Rebaba's

unique cooling system. Visible certifications could also help instil security in the customer by making the product feel more legitimate. The new EU directive regarding battery passports will make it easier for customers to get product specification information. In addition, Rebaba could also present information comparing 1st life and second-life batteries on their website or digital communication to reduce confusion.

When it comes to BESS there are a lot of benefits to using second-life batteries, not only is it cheaper but it's also more sustainable environmentally and socially. Many industries and commercial facilities strive to become greener and reduce environmental impact. The benefits of using second-life electrical vehicle batteries should be communicated to make customers satisfied with their choice. Having visible signage indicating that Rebaba's BESS uses second-life batteries adds value for customers because it signals a more sustainable investment.

The Design for Exchange strategy highlights how products may be obtained by one user and cleared out by another. It is important to account for the different arising needs of the current and future users. Future users might need to properly assess the product they will be reusing and evaluate its value. Such is the case for the reused electric vehicle batteries used in the BESS. One way to make customers feel more comfortable purchasing a product with reused parts is by allowing them to try the product before purchase. In the case of a BESS, this could be done by inviting potential customers to see previously successful projects. Designing for exchange in a product like a BESS differs somewhat from other products since it is a relatively complex system with many cooperating parts. This design strategy prompted discussions regarding the transportation of the batteries and that it would save time in the installation if the batteries were transported inside the cabinets. It was further discussed that clear instructions and checklists would make it easier to assemble and disassemble the system and thus make it easier to exchange smaller components if needed.

Another way to facilitate the circularity of products is to design for match-making, making it easier for users to connect and interact to exchange products. An idea discussed was the possibility of having a Rebaba community where customers can interact and trade components and information regarding the system. More ways to match a product to a user is to inform about the differences between the Rebaba offering and its competitors. A show-room could in the same way be used to

highlight the benefits of Rebaba's BESS and make them stand out. How the company communicates its offering impacts finding customers that can benefit from Rebabas system.

4.4 Concept Generation

Ideas from the ideation workshop and circular design ideation were combined to form various concepts. The concepts were defined with sketches and text and developed using the function analysis presented in section *3.5 Function Analysis* to identify vital and desirable functions. Emphasis was put on contributing functions to better ergonomics and adapting the solution to the customer group. Three concepts *Trust*, *Involve*, and *Value Maintenance* were created to tackle separate parts of the product-service timeline.

4.4.1 Trust Concept

This concept aims to tackle customers' concerns regarding the safety of BESS and second-life batteries. This is done by incorporating various elements in the company website that provide trust and clarity before and during the BESS project. Before initiating a project Rebaba would offer an evaluation where they assess the customers' needs and conditions to recommend a suitable solution. The service will be transparent and honest and not require proceeding with the project. This will reduce the threshold for customers to initially contact Rebaba and establish the company as a legitimate and benevolent integrator.

Also, the concept provides a design feature where customers can design their own BESS based on parameters eg. output power, intended use and placement. The website would also contain informed about Rebaba's solution, the benefits of using second-life batteries, and videos showing how the safety system would manage a battery that undergoes thermal runaway. Customers may feel safer buying the product when convinced that the system can handle the worst-case scenario. Showcasing earlier successful projects and customer reviews will also be utilised to build credibility.

When customers choose to start a project with Rebaba they would receive an account for a portal accessed through the website. The account is used to follow active projects and collect relevant

information. Rebaba will coordinate third-party companies through the portal to eg. build the foundation. However, Rebaba would always carry the conversation with the customer to maintain a strong relationship with the customer and further establish them as a legitimate integrator. Figure 49 shows how the Rebaba website could be designed to incorporate this concept.



Figure 49. A sketch of the Rebaba website.

4.4.2 Involve Concept

This concept intends to make people feel secure with BESS by normalising interaction with the system. It is specifically directed at commercial facilities with limited space, where an option could be to place the BESS in a visible place such as a lobby. The BESS will be wrapped in the art of the customer's choice and feature an interface on the cabinet. The artwork is intended to make people curious and encourage them to get closer to the BESS. Information about the artwork will be visible in the interface together with other relevant information and statistics about the performance of the BESS and second-life batteries. Inside the cabinet, there will be a second interface that can only be accessed by authorised personnel. The inner interface will have different

modes that customise the interface to the specific user eg. customer or service personnel. Figure 50 shows a sketch of the concept.

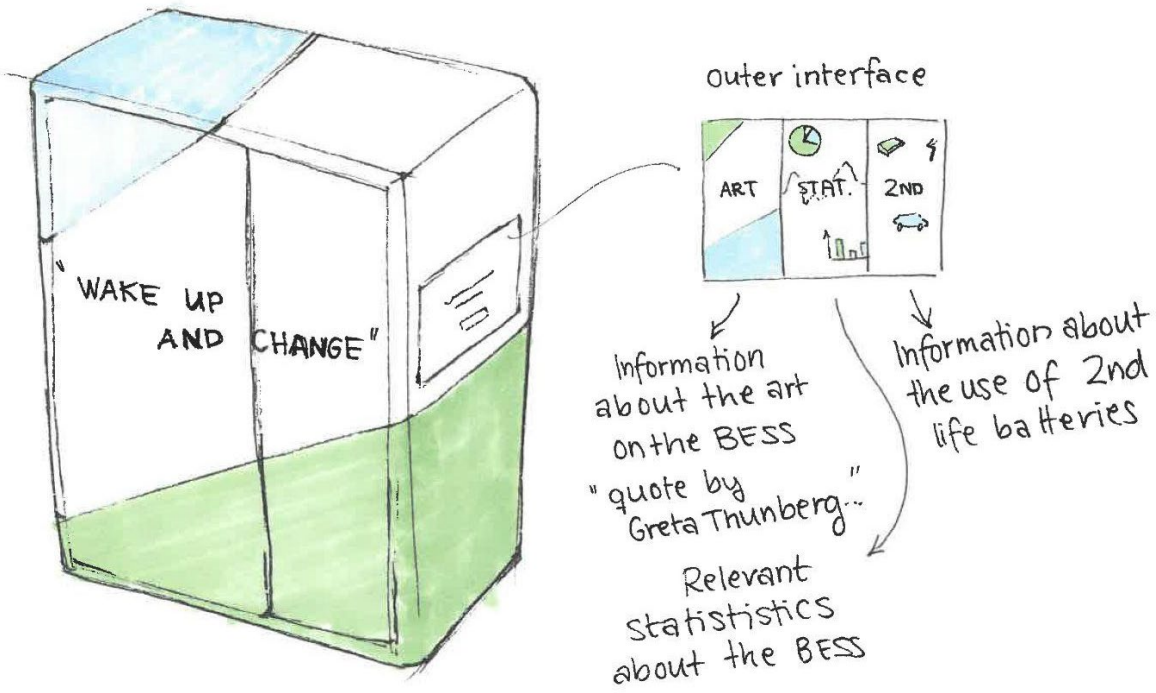


Figure 50. Shows the BESS depicting an artwork with an interface.

4.4.3 Value Maintenance Concept

This concept was created to maintain the value of the systems after installation. This will be done by regular maintenance by service personnel and by encouraging the customer to perform regular checks of the system's health. The concept features a Rebaba portal where customers can see the status and find information on how to take care of their system. Notifications will also appear in the portal when it is time for servicing. Additionally, customers would access a Rebaba community where they can discuss questions or issues with others with more knowledge. The community serves as an added layer of support to hinder the off-time of the BESS. Figure 51 shows a sketch of the Rebaba Posrtal that would maintain the value of the BESS after installation.

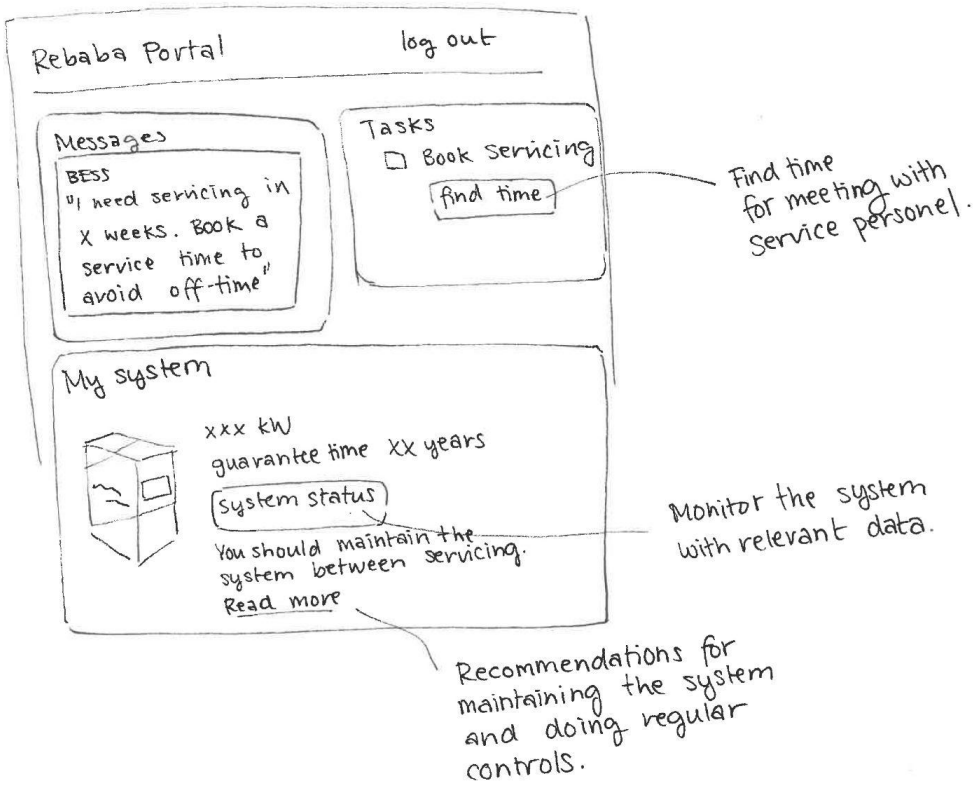


Figure 51. Showing how the value of the system would be maintained with the help of the Rebaba Portal.

4.5 Concept Selection

The concepts were discussed based on the insights and the impact the solution could have. It was decided that the final design proposal would combine all three concepts.

The *Trust* concept was appealing since it targets worries regarding second-life lithium-ion batteries presented in insight card five. This insight is important to target since people need to feel comfortable with the technology to invest in a BESS. Some customers refrain from solutions that do not communicate well.

The *Involve* concept had many advantages when it came to the limited space of C&I however for this concept to be effective it was still important to tackle the issues that the *Trust* concept did. This concept aims to encourage placement indoors and close to people. Part of this concept gives

customers the option to change the appearance of the BESS to promote the product being placed in visible places.

The *Value Maintenance* concept provides a solution to the desire to maintain the value of a service. The solution helps maintain the quality of the BESS through time and makes a better customer experience when handling the BESS. Also, BESS projects are complicated and being able to provide a platform supporting the projecting was considered important.

The opportunity to incorporate these concepts in the Rebaba Website made it easier to combine the different concepts.

4.6 Concept Development

The concept was developed using the digital prototyping tool Figma. The work followed an iterative process where various parts of the solution were redesigned multiple times. Inspiration was taken from websites such as Patagonia, Polarium and Tesla. Inspiration was also drawn from the automobile sector and how the products are presented on their websites. When developing the final solution, it was important to create intentional features that serve a purpose and continuously think of the user experience.

Chapter 5

Deliver

This chapter presents the final design proposal. The solution consists of two design solutions, the Rebaba Website and the Rebaba portal, which combine ideas from previously discussed ideas. The two design solutions tackle different problems.

5.1 Rebaba Website

The Rebaba Website was designed to aid in introducing BESS using second-life EV batteries to customers and initiate comfortable and reliable contact with Rebaba. The first page of the website provides comprehensive information about the company and its offerings. When entering the website they are greeted with the company tagline “Empowering Flexible Energy Solutions” sparking interest. Figure 52 shows the first page of the website with the company tagline.

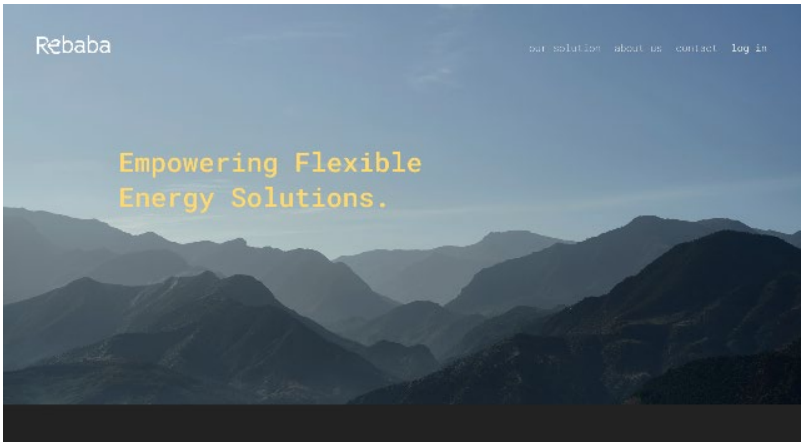


Figure 52. The first page of the website greets the visitor with the company tagline.

Below, the page presents the BESS offering and how it differs from others on the market. It emphasises the circularity of the product and their thoughtful solution for keeping the system safe to quickly attend to customers’ worries. Beneath, the full-service approach is highlighted in a three-part illustration describing the main parts of their product-service solution: analysis, delivery, and maintenance. It aims to convey the company’s desire to support the customer as a part of their offering. Rebaba will work to collaborate with the customer to achieve a well-adapted solution. Their transparent approach through truthfully analysing the customer needs and consistently assisting with maintaining the system aligns with the brand words benevolent and reliable presented in *2.4.2 Brand in Words*. Figure 53 shows the introduction to the Rebaba offering.

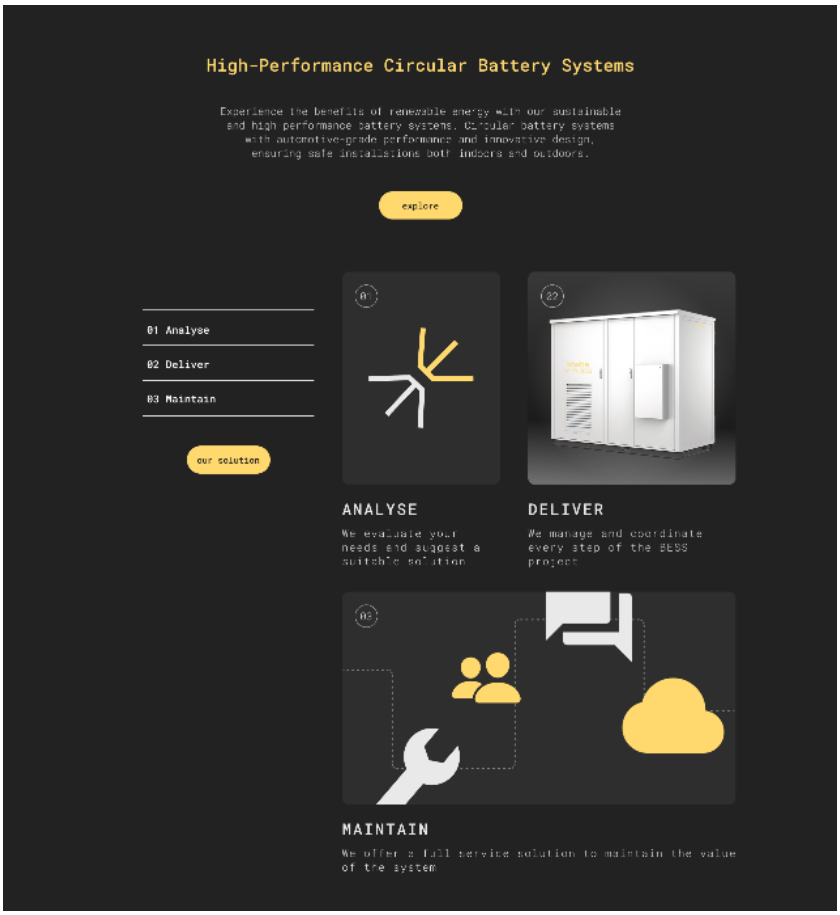


Figure 53. Shows the introduction to the Rebaba offering on the first page.

Further down the page is a section called *Latest Stories* highlighting the Rebaba communication. This section presents news in the field, blog posts written by the company or educational material about the topic intensifying their present character. It enhances the brand objective to be collaborative and contribute to progress in the area by involving and educating others. Being transparent about the topic, and informing about the solution and field, builds trust in the brands' expertise. Figure 54 shows the display of communication on the first page.

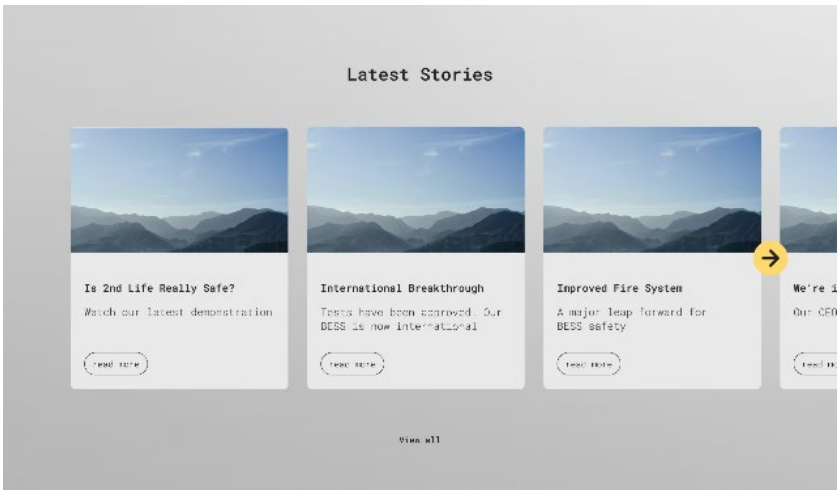


Figure 54. Communication through the Latest Stories on the first page.

The section *Our Achievements* is dedicated to showcasing earlier projects and reviews. It features pictures from the site, a quote from the client, and their rating of Rebabas services. This is intended to yield security among new clients by showcasing previous projects. It provides tangible information about the real-life delivery of BESS and an understanding of the possibilities of the solution. Figure 55 shows the lower part of the home page.

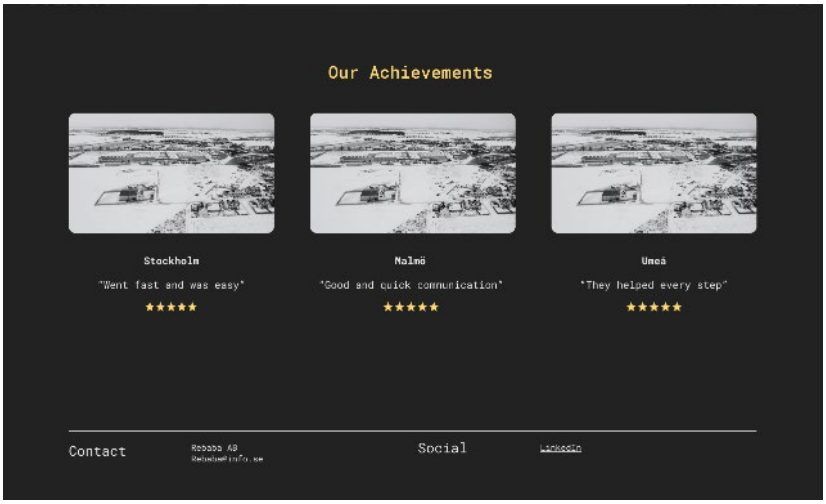


Figure 55. shows the lower part of the home page including the Latest Stories and Our Achievements.

5.1.1 Product Offering

It is important to highlight the benefits of the Rebaba solution to customers. The page *Our Solution* provides in-depth information about the Rebaba offering in two main parts: *product* and *safety*. Figure 56 shows the introductory first part of the *Our Solution* page, describing the BESS systems and their benefits.

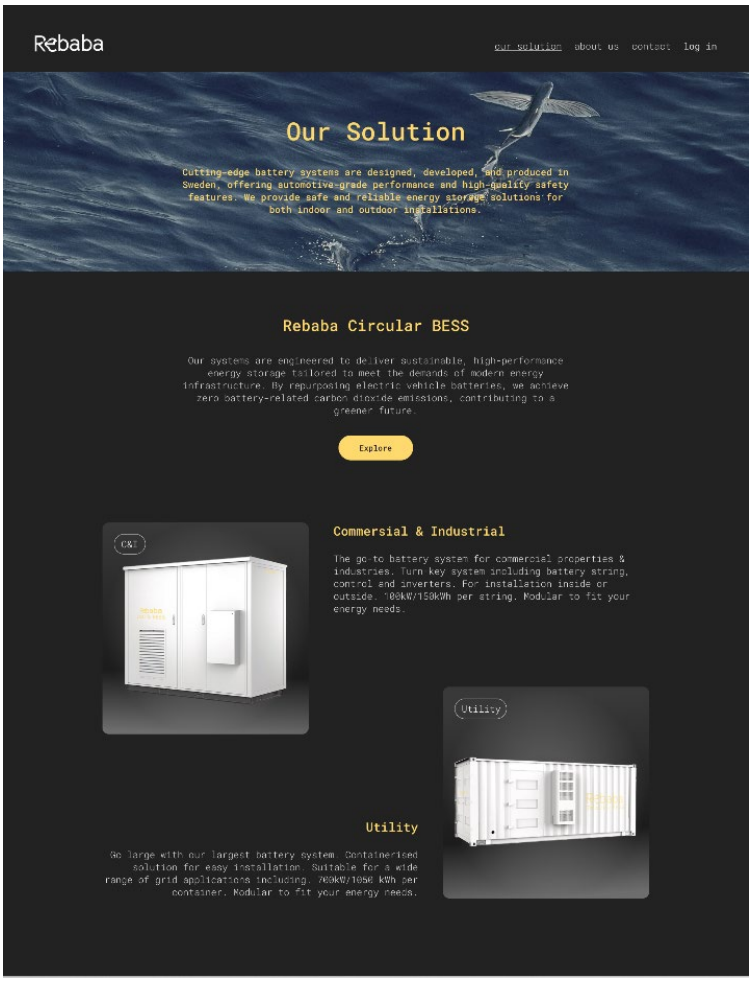


Figure 56. The first part of the Our Solution page describes the BESS solution.

The Rebaba solution is distinguished by its cooling and safety solution that enables great system flexibility. The second part of the *Our Solution* page offers in-depth information about the safety benefits of the solution. It informs the visitor about the system tests and a video showing the testing. This makes customers feel secure with the technology and validates its legitimacy. Finally, the total offering is again shown to inform users of Rebaba's presence throughout the whole way,

contributing to trust in the company. Figure 57 shows the safety information on the lower part of the *Our Solution* page.



Figure 57. The second part of the Our Solution page with information about safety, testing of the system and the total offering.

5.1.2 Configurator

The website allows users to play with the configuration of the BESS without any requirement to proceed further. Like vehicle companies, the BESS can be configured digitally to explore its features and familiarise with the product. When doing so the user chooses the desired output power, intended use, placement and appearance which modifies the displayed system. The customer gets a date for possible delivery, an approximate price for the configurated BESS, and the possibility to initiate contact with Rebaba for a quote. The page also highlights a few of the

most frequently asked questions and answers to help inform the visitor about the solution. Figure 58 shows the configurator feature where users can configure their system.

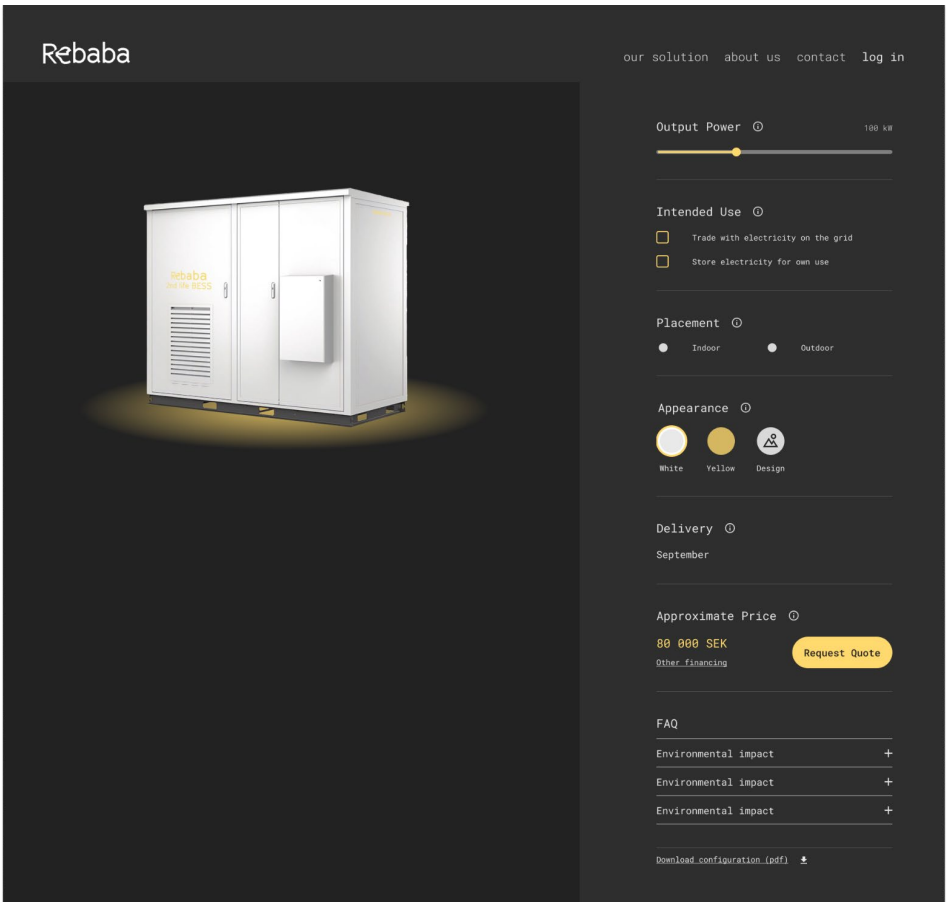


Figure 58. Configuration feature

5.2 Rebaba Portal

A web portal was designed to help manage BESS projects and systems. Many parties are involved in delivering a BESS which complicates the project. If Rebaba for instance has contracted a third-party company to build the foundation of a BESS, any information relevant to the customer should be relayed through the Rebaba Portal. A portal enables clear interaction between the customer and Rebaba, ensuring a strong relationship with the customer and that Rebaba is perceived as a reliable company caring for its customers.

When a BESS project has been initiated the customer gets an account on the Rebaba Portal where the project is managed and monitored. The account can be accessed by several employees in an organisation so everyone involved can stay updated. Through the portal, customers also access information and data about the systems in service. The Rebaba Portal was designed to make the management of BESS projects smoother by clearly distributing tasks and collecting information in one place. The portal is also a tool to maintain the system's value with encouragement for serving and information for maintenance. Figure 59 shows the log-in frame for the Rebaba Portal with information for accessing it.

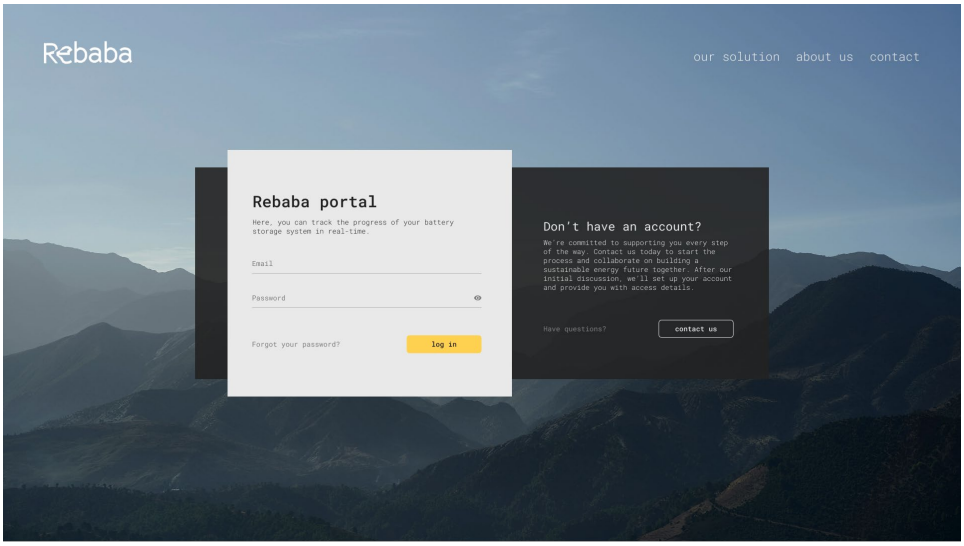


Figure 59. Shows the login for the Rebaba Portal.

When users log into the Rebaba portal they are presented with the *Home* page where an overview of important data and notices regarding their ongoing BESS project and operating systems is presented. The page features a calendar providing an overview of active projects and tasks, and a display of notifications informing the user of new tasks or other communication by the team. The page also previews the users' products with quick links to their separate pages. To support the user the *Home* page also supplies different ways to assist the user with a chat function and a link to the *Rebaba Community*. This lowers the required effort for seeking assistance by accessing it easily. The page also provides an overview of the data collected from the systems visualised in a graph to compare the data over time. The portal is designed with dark colours to decrease energy consumption when displayed. Figure 60 shows the start page of the Rebaba portal.

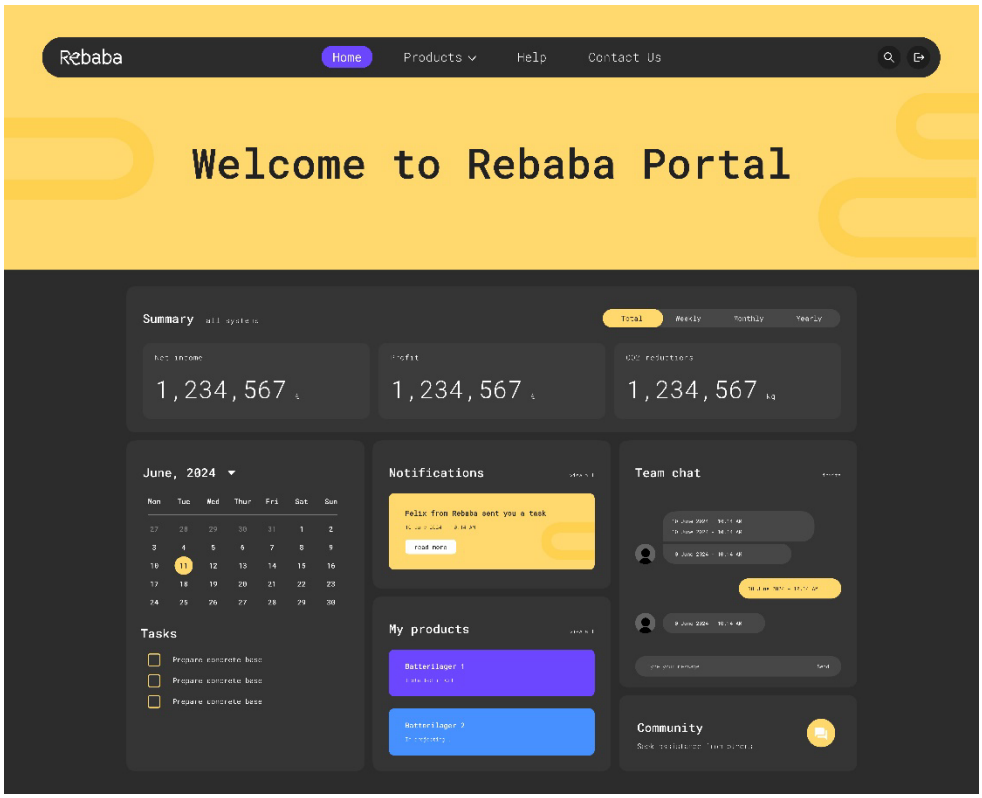


Figure 60. The Start page of the Rebaba Portal.

Each associated page can be accessed through the widgets or the top menu. The menu is constantly visible on top of each page indicating what page is active, providing links to the other pages, and the options to search the page and to sign out of the portal. The active page is highlighted in purple to clearly indicate the system status to the user. Figure 61 shows the top menu of the Rebaba portal.



Figure 61. The top menu of the Rebaba Portal

5.2.1 Products

The *Products* page contains information about the users' products. All systems both in progress and operating are monitored through the platform and found through the *Products* page. The page provides an overview of all systems each summarised in their widget. Figure 62 shows the overview page for a customer that has two systems one in progress and one operating.

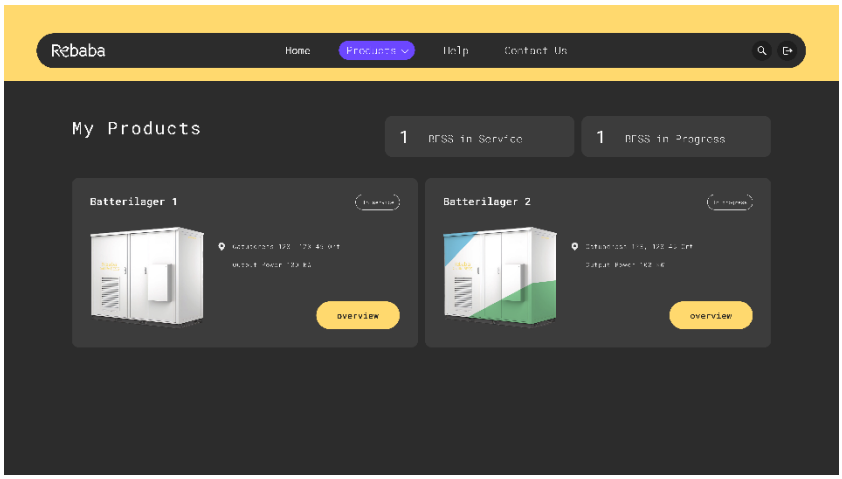


Figure 62. Products preview page.

Each system page is divided into three tabs: *Overview, Statistics,* and *Maintenance*. By clicking a widget the user navigates to the individual system *Overview* page containing more information about the system. A system that is in the process of being delivered the information on the page will attend to the project. The page contains a specification of the system, information regarding the project status and tasks for completion of the project. Instructions by Rebaba will be directed through the *Notice* section meant to inform customers of project-specific tasks and how to perform them. Figure 63 shows the system in process *Overview* page.

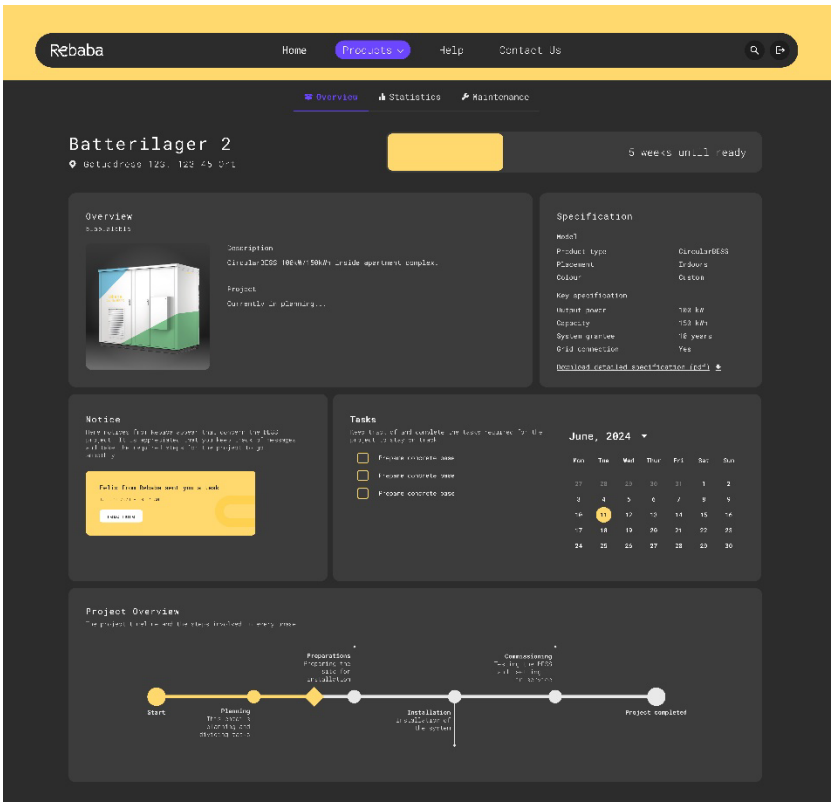


Figure 63. Overview page of a system in the process.

A system that is operating is monitored through the portal. The page provides relevant data regarding the system generated eg. income, profit, and carbon dioxide emission savings. Detailed data about the system is visualised under the *Statistics* tab. The *Overview* page also communicates the system's maintenance needs and reminds the customer to care for the system. Encouraging regular maintenance of the system ensures prolonged use and reinforces confident handling and reduces feelings of unease. Figure 64 shows the *Overview* page of an operating system.

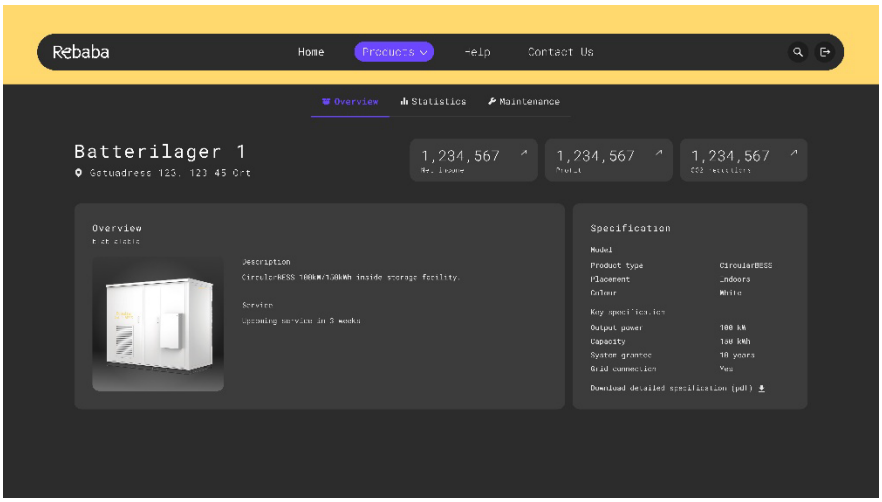


Figure 64. Overview page of an operating system.

The *Maintenance* tab of the system page provides in-depth maintenance information. Most noticeably users can first see a yellow widget that contains information regarding the time since the last servicing and information regarding upcoming service. Regular maintenance and servicing are encouraged to maintain the value of the system, making this feature very important. The feature is therefore highlighted in yellow making it more noticeable. The page also provides a *Servicing log* displaying the history of service visits similar to that of a car. It also teaches and recommends ways to maintain the system. The recommendations can be printed and placed beside the BESS to encourage users to regularly check the system's well-being. Figure 65 showcases the *Maintenance* tab.

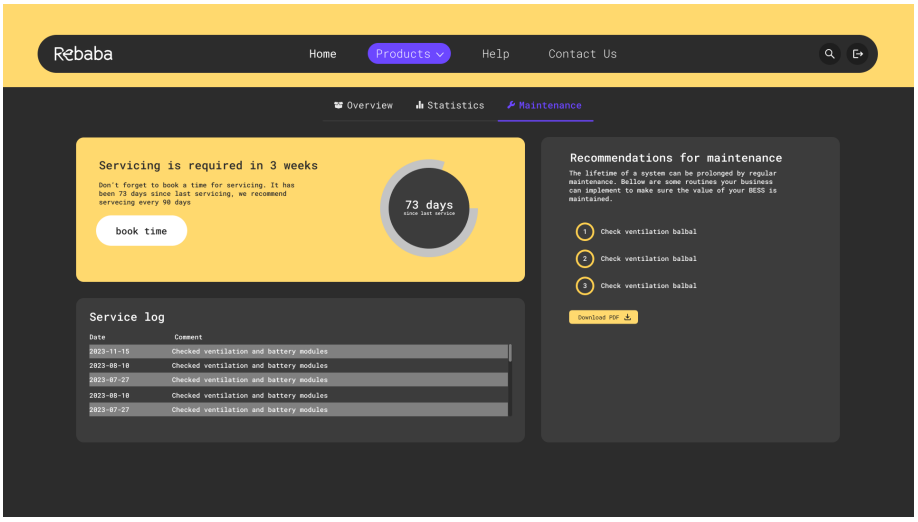


Figure 65. The Maintenance page.

5.2.2 Help

The *Help* page accessed through the top menu contains information to assist the user. It offers three service options: *Community, Guides,* and *FAQ*. Each option provide different ways of supporting the user. The user is also given the option to search the help center using key words in the search bar. Figure 66 shows the *Help* page and the three options.

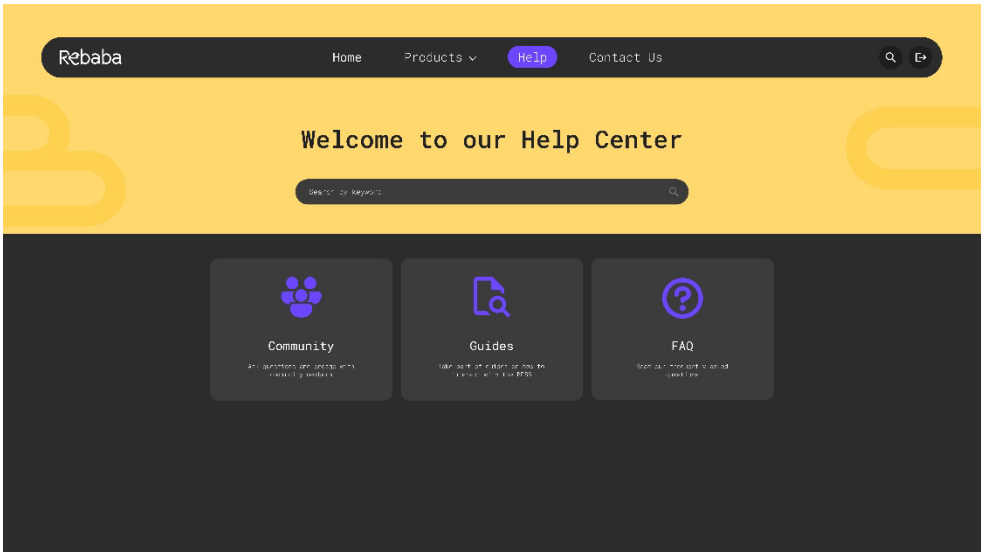


Figure 66. The Help Center page.

The *Community* option works to connect Rebaba customers. It is a space where users create posts, ask questions, share thoughts and discuss topics. The community is a way for users to share information to optimise and simplify the use of the systems. The idea is that Rebaba create a transparent platform that genuinely encourage learning. Experienced users with more knowledge such as utility users can also help others with less knowledge and resources such as C&I. The design of the Rebaba Community was inspired by the *Quora* platform and features a menu on the left where users can navigate to the main feed, written posts, linked ports and saved posts. Figure 67 shows the *Rebaba Community* page.

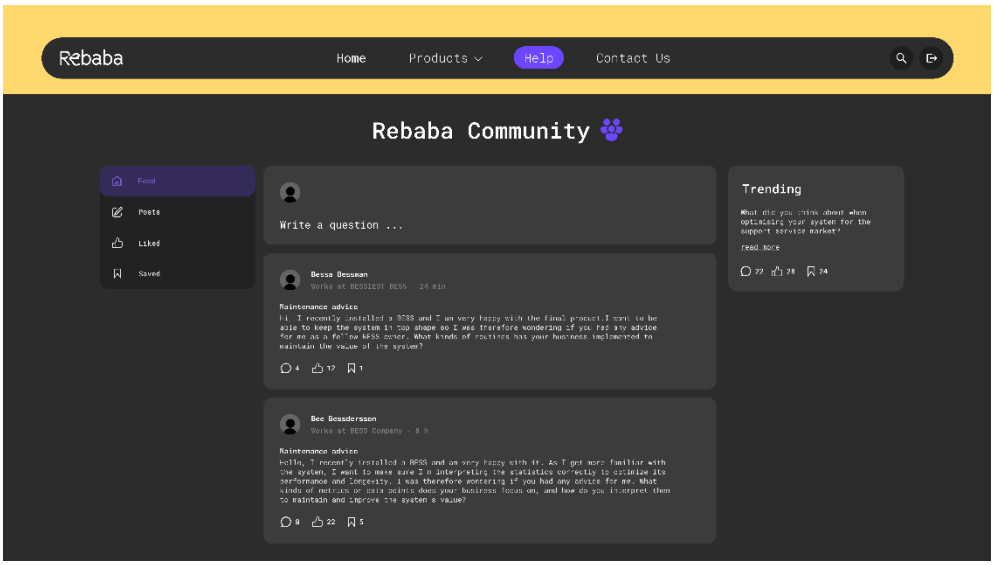


Figure 67. The Rebaba Community page.

The *Guides* option contains guidelines designed to assist customers during the project management, operation and safety practices of the BESS. It features a title, and a short description of the guideline and allows users to download a PDF with more information. The guidelines that RISE developed are also available for download at the bottom of the page. Figure 68 shows the *Guidelines* page.

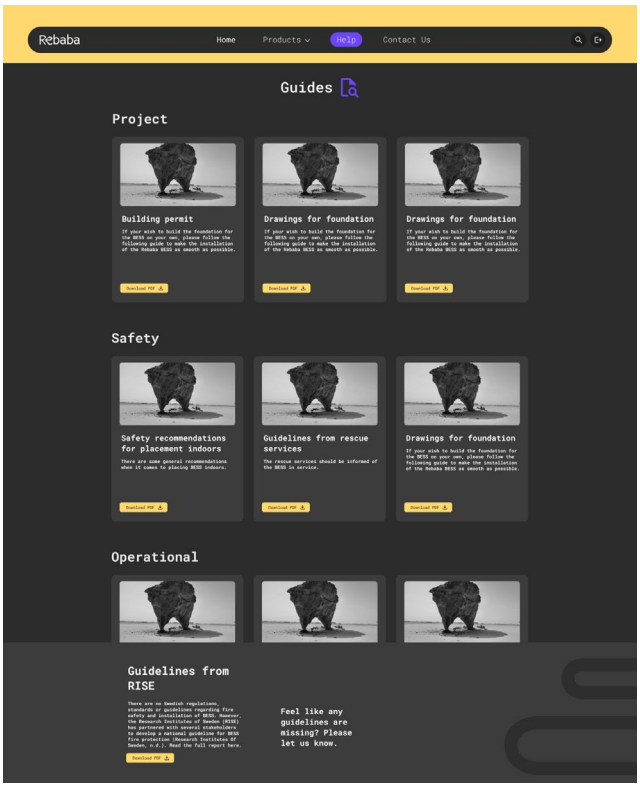


Figure 68. The Guidelines page.

Finally, the *FAQ* option contains the most frequently asked questions about the solution and answers. The questions are categorised according to the topic. Figure 69 shows the *Frequently Asked Questions* page.

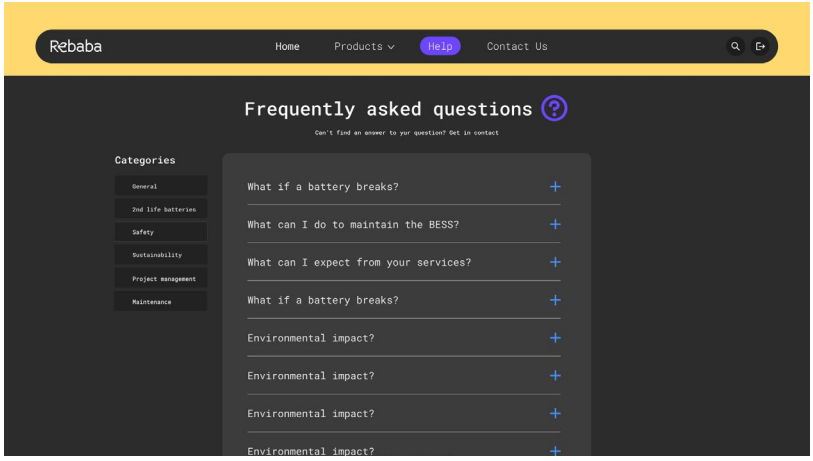


Figure 69. The Frequently Asked Questions page.

Chapter 6

Discussion

In this chapter the final concept is evaluated and its potential future improvements is discussed.

6.1 Evaluation

The concept could be evaluated further to determine the effectiveness of the proposed design solution. It would be interesting to set up a usability test where participants are instructed to perform various tasks on the website and see how comprehensible the features and layout are. A think-out-loud method would be beneficial for the test participants to voice their opinions. Furthermore, it would be interesting to test the Rebaba portal during a BESS project to see how it affects the work and organisational ergonomics. It would be interesting to compare the final concept with existing project management and maintenance tools. This could be a valuable way of evaluating the concept as well as identifying things that might be missing.

A potential challenge with the final solution could be implementing the Rebaba portal for the management of the BESS projects. The ease of adoption needs to be evaluated and training might be required for Rebaba staff to use the platform. Strategies for smooth onboarding and continued use should thus be explored.

The final concept should also be evaluated from a feasibility standpoint. It should be investigated how much time and resources it would take to realise the website and online platform for someone with coding knowledge. The concepts were created based on real-life examples which hopefully would entail that the concepts are possible to develop.

6.2 Limitations

The project has a few limitations, including the diversity in the gender of the interviewees. It would have been beneficial to come in contact with more women in the profession to get a more accurate picture of their experiences with BESS. Even though some interviews were conducted with installers, it would also be interesting to have had more interviews and contact with several. Finally, it would be rewarding for the project to have seen and interacted with the systems that are currently used when managing the projects, monitoring and maintaining BESS.

6.3 Recommendations for future work

The Rebaba *Community* could be further developed. The *Community* could be utilised as an additional resource to maintain the value of BESS. It could include third-party companies and customers willing and knowledgeable to help with repairs. In the case of a repair being needed in the countryside or more remote areas, such a feature would make it possible for Rebaba to provide servicing more spread out. To give users an incentive to interact, a reward system could be implemented where customers who help other customers are rewarded with a discount on their service agreement or future systems. Further consideration is also needed for managing and monitoring the activity on the *Community* page.

The impact of the *Our Achievements* feature on the Rebaba webpage could be reinforced by allowing potential customers to reach out to current customers to discuss the reviews further. Additionally, it could also be reassuring for potential customers if they could visit the sites presented in *Our Achievements*.

The strength of this concept is that it is digital and can be adjusted to evolving needs. However, Rebaba would need to have an entire team dedicated to running the website and platform to evolve it continuously.

6.4 Conclusion

This thesis project aimed to explore how the ergonomics and usability of Battery energy storge systems could be improved. Literature research, together with in-depth interviews and study visits were conducted to understand the subject and find areas that needed improvement. Insights from the research showed that the cognitive ergonomics and organisational ergonomics of the project management of BESS could be improved due to projects being complex and tailored to each customer. It was also found that customers value the service that the BESS provides more than the physical product. A hindrance specific to the BESS using second-life electric vehicle batteries is the concern among customers regarding safety. Based on the insights a website and project management platform were developed to aid in the understanding od these systems, manage the

project and maintain the product's value over time. The final concept has areas that could be improved by evaluating the concept and focusing on the design of specific features.

8. References

Printed Referenences

Dash, P., & Hu, Y. C. (2021, june 24). *How much battery does dark mode save?: an accurate OLED display power profiler for modern smartphones*. ACM Digital Library. Retrieved May 29, 2024, from https://dl.acm.org/doi/10.1145/3458864.3467682

Deng, D. (2015). *Li-ion batteries: basics, progress, and challenges*. https://onlinelibrary.wiley.com/doi/full/10.1002/ese3.95

Dong, Q., Liang, S., & Li, J. (2023). *Cost, energy, and carbon footprint benefits of second-life electric vehicle battery use*. ScienceDirect. https://www.sciencedirect.com/science/article/pii/S2589004223012725

Grönlund, O., Quant, M., Rasmussen, M., Willstrand, O., Hynynen, J., & Rise Research Institute of Sweden. (2023). *Guidelines for the fire protection of battery energy storage systems*. https://ri.diva-portal.org/smash/record.jsf?pid=diva2%3A1824438&dswid=7569

Hansons Tech. (2023, june 28). *LED Computers: The Future of Visual Computing, Introduction, History, Advantages & Disadvantages and Conclusion.* Medium. Retrieved May 29, 2024, from https://medium.com/@hasonsnik/led-computers-the-future-of-visual-computing-74c3ed2dd32e

IEC International standard. (2020). *Electrical energy storage (EES) systems – Part 5-2: Safety requirements for grid-integrated EES systems – Electrochemical-based systems*.

Official Journal of the European Union. (2023). *REGULATION (EU) 2023/1542 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL*.

Research Institutes Of Sweden. (n.d.). *Vägledning brandskydd för batterienergilager*. ri.se. https://www.ri.se/sv/vad-vi-gor/projekt/vagledning-brandskydd-for-batterienergilager

Rose, L., & Eklund, J. (2019). *ERGONOMI I PRODUKT- OCH PRODUKTIONSUTVECKLING*.

Surtini, R. (2021, May 6). *Flywheel Energy Storage*. Science Direct. Retrieved February 28, 2024, from https://www.sciencedirect.com/topics/engineering/flywheel-energy-storage

Other references

Amnesty International. (n.d.). *MÄNSKLIGA RÄTTIGHETER OCH MILJÖ I BATTERIVÄRDEKEDJAN*. amnesty.se. https://www.amnesty.se/vara-rattighetsfragor/foretagsansvar/hallbara-litiumjonbatterier/

Blanace Power. (2023, September). *What Are The Best Types Of Battery Energy Storage Systems*. BalancePower. https://balancepower.co.uk/news-insights/what-are-the-best-types-of-battery-energy-storage-systems

CATL. (n.d.). *Storing Infinite Energy Energy Storage System Solutions and Products*. https://www.catl.com/en/uploads/1/file/public/202303/20230315092000_ahw9vpn63j.pdf

CATL. (2022, May 11). *CATL's EnerOne battery storage system won ees AWARD 2022*. https://www.catl.com/en/news/935.html. catl.com

CellBlock FCS. (2022, October 4). *Lithium-ion Thermal Runaway Test, in CellBlock FCS SUPERMAX Case*. Youtube. https://www.youtube.com/watch?v=Axlrx3qoIic&ab_channel=CellBlockFCS

Comsys. (n.d.). *ACTIVE DYNAMIC FILTER ADF P100*. comsys.se

Dahlqvist, J. (2024, april 26). *Rebaba återvinner elbilsbatterier* [Audio podcast episode]. Rapidus. https://open.spotify.com/episode/32NB18UtIcAQMCg5b7l34h?si=J_KkE967QFaG9TOIsCbopw

Design Council. (n.d.). *The Double Diamond*. Design Council. https://www.designcouncil.org.uk/our-resources/the-double-diamond/

Ecactus Solar UK. (n.d.). *CATL EnerC*. eCactus Solar. Retrieved April 18, 2024, from https://ecactussolar.co.uk/products/catl-enerc/

Energiföretagen. (2023, April 6). *Effektbalansen i Sverige - Elproduktion*. Energiföretagen Sverige. Retrieved March 4, 2024, from https://www.energiforetagen.se/energifakta/elsystemet/produktion/effektbalansen/

Energiföretagen Sverige. (2022, January 12). *Planerbart, energilager, kundflexibilitet – Energiföretagen förklarar*. Energiföretagen Sverige. Retrieved February 22, 2024, from https://www.energiforetagen.se/pressrum/nyheter/2022/januari/planerbart-energilager-kundflexibilitet--energiforetagen-forklarar/

EVBox. (2023, May 16). *What are electric car batteries made of? | EVBox*. EVBox. Retrieved February 6, 2024, from https://blog.evbox.com/what-are-ev-batteries-made-of

Evesco. (n.d.). *Battery Energy Storage System Components*. Power Sonic. power-sonic.com

EVKX. (2024, April 28). *Battery pack & configuration :: evkx.net*. EVKX.net. Retrieved May 21, 2024, from https://evkx.net/technology/battery/batterypack/

GlobalData. (n.d.). *Contemporary Amperex Technology Ltd: Overview*. globaldata.com. https://www.globaldata.com/company-profile/contemporary-amperex-technology-ltd/

GlobalData Thematic Intelligence. (2023, October 30). *The cost of green energy: lithium mining's impact on nature and people*. Mining Technology. https://www.mining-technology.com/analyst-comment/lithium-mining-negative-environmental-impact/?cf-view&cf-closed

gridX. (2024, January 29). *Peak Shaving*. gridX. https://www.gridx.ai/knowledge/peak-shaving

Guerra, M. (2023, November 14). *Second-Life EV Batteries: Benefits, Challenges, and Innovations*. Battery Technology. https://www.batterytechonline.com/ev-batteries/second-life-ev-batteries-benefits-challenges-and-innovations

Jolly, J. (2023, December 1). How problematic is mineral mining for electric cars? *The Guardian*. https://www.theguardian.com/business/2023/dec/01/do-electric-cars-have-problem-mining-for-minerals

Lee, C. (2023, June 5). *Comparison Between Three Types of Lithium Ion Battery Cell - Cylindrical Cell, Prismatic Cell, Pouch Cell*. LinkedIn. Retrieved February 6, 2024, from https://www.linkedin.com/pulse/comparison-between-three-types-lithium-ion-battery-cell-cliff-lee/

MacLead, D. (2013, September 8). *10 Principles of Cognitive Design – Ergoweb LLC*. Ergoweb. Retrieved March 6, 2024, from https://ergoweb.com/10-principles-of-cognitive-design/

Mahajan, S. (2017, November 9). *Is Dark mode a Greener mode?* YouTube: Home. Retrieved May 31, 2024, from https://www.linkedin.com/pulse/dark-mode-greener-suyash-mahajan/

Maisch, M. (2023, December 1). *CATL: Staying on top of the battery game*. pv magazine. https://www.pv-magazine.com/2023/12/01/catl-staying-on-top-of-the-battery-game/

McKinsey & Company. (2023, August 2). *Enabling renewable energy with battery energy storage systems*. mckinsey. https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/enabling-renewable-energy-with-battery-energy-storage-systems

Minos, S. (2023). *How Lithium-ion Batteries Work.* energy.gov. https://www.energy.gov/energysaver/articles/how-lithium-ion-batteries-work

Mordor Intelligence. (n.d.). *Battery Energy Storage System Market Size & Share Analysis - Growth Trends & Forecasts (2024 - 2029) Source: https://www.mordorintelligence.com/industry-reports/battery-energy-storage-system-market*.

Murray, C. (2024, April 15). CATL unveils 'five-year zero degradation' BESS with 6.25MWh per container. *Energy Storage News*. https://www.energy-storage.news/catl-unveils-five-year-zero-degradation-bess-with-6-25mwh-per-container/

Naturvårdsverket. (n.d.). *Producers responsibility for batteries*. naturvardsverket.se. https://www.naturvardsverket.se/en/guidance/extended-producer-responsibility-epr/producers-responsibility-for-batteries/#E97562860

Nielsen, J. (1994, April 24). *10 Usability Heuristics for User Interface Design*. Nielsen Norman Group. Retrieved May 29, 2024, from https://www.nngroup.com/articles/ten-usability-heuristics/

Olsson Jeffrey, M. (2023, October 5). *Med Rebaba tar Paula Runsten vara på elbilars batterier*. Dagens industri. Retrieved February 6, 2024, from https://www.di.se/digital/hon-vill-ge-uttjanta-elbilsbatterier-nytt-liv/

Polisen. (2022, february 1). *Trafikregler för polisen | Polismyndigheten*. Polisen. Retrieved May 21, 2024, from https://polisen.se/lagar-och-regler/trafik-och-fordon/regler-for-polisen-i-trafiken/

Quantum Scape. (2021). *Distinguishing charge rates for next-generation batteries*. QuantumScape. https://www.quantumscape.com/resources/blog/distinguishing-charge-rates-for-next-generation-batteries/

Senior, R. (2022). *Ideal Operating Temperatures for Lithium Batteries*. Fortress Power. https://www.fortresspower.com/ideal-operating-temperatures-for-lithium-batteries/

Siemens Energy. (n.d.). *Battery Energy Storage | BESS*. Siemens Energy. Retrieved February 6, 2024, from https://www.siemens-energy.com/global/en/home/products-services/product/battery-energy-storage.html

Singh, B. (n.d.). *Dark Mode: Accessibility vs Sustainable Web Design* -. CueBlocks. Retrieved May 29, 2024, from https://www.cueforgood.com/blog/dark-mode-accessibility-sustainable-web-design/

Sudhakar, N. (2023). *UNDERSTANDING THE NEW EU BATTERY REGULATION Key Provisions and Impact of the New EU Battery Regulatory Explained*. tuvsud.om. https://www.tuvsud.com/en-us/resource-centre/blogs/mobility-and-automotive/understanding-the-new-eu-battery-regulation

Svenska krafnät. (2024, February 19). *Om olika energikällor*. Svenska kraftnät. Retrieved February 21, 2024, from https://www.svk.se/om-kraftsystemet/om-olika-energikallor/

Svenska kraftnät. (2021, March 9). *Översikt av kraftsystemet*. Svenska kraftnät. Retrieved February 21, 2024, from https://www.svk.se/om-kraftsystemet/oversikt-av-kraftsystemet/

Svenska kraftnät. (2022, December 8). *Sveriges elnät*. Svenska kraftnät. Retrieved February 21, 2024, from https://www.svk.se/om-kraftsystemet/oversikt-av-kraftsystemet/sveriges-elnat/

Svenska kraftnät. (2023, December 21). *Energilagring med batterier och vätgas*. Svenska kraftnät. Retrieved February 21, 2024, from https://www.svk.se/om-kraftsystemet/energilagring-med-batterier-och-vatgas/

Swedish Chemicals Agency. (2024, February 6). *Batteries*. Kemi.se. https://www.kemi.se/en/rules-and-regulations/additional-eu-rules/batteries

Swedish Work Environment Authority. (2020). Swedish Work Environment Authority. https://www.av.se/en/health-and-safety/work-posture-and-workload--ergonomics/

This Is Service Design Doing. (n.d.). *DEVELOPING KEY INSIGHTS*. thisisservicedesigndoing. thisisservicedesigndoing.com

Tung, L. (2018, November 9). *Google: Here's why dark mode massively extends your OLED phone's battery life*. ZDNET. Retrieved May 29, 2024, from https://www.zdnet.com/article/google-heres-why-dark-mode-massively-extends-your-oled-phones-battery-life/

UL Solutions. (2019, December 2). *UL 9540A Battery Energy Storage System (ESS) Test Method*. ul.com. https://www.ul.com/news/ul-9540a-battery-energy-storage-system-ess-test-method

United Nations. (n.d.). *THE 17 GOALS*. sdgs.un.org. https://sdgs.un.org/goals/goal12

Use2Use. (n.d.). *Design opportunities*. USE2USE. Retrieved May 2, 2024, from https://www.use2use.se/opportunities-for-designers/

Vätgas Sverige. (n.d.). *Energilagring*. Vätgas Sverige. Retrieved March 4, 2024, from https://vatgas.se/fakta/energi-lagring/

Wu, B. (2023). *Top 10 Battery Energy Storage System Companies*. LinkedIn. https://www.linkedin.com/pulse/top-10-battery-energy-storage-system-companies-ben-wu/

Appendix A - Stakeholder Map

The stakeholder map seen in Figure 1 shows the stakeholders involved in the Battery Energy Storage System that Rebaba will deliver to customers. The map is centred around the user and the stakeholders are primary or secondary. A package symbolises goods or services, the dollar sign stands for money, the person symbolises labour, the sheet of paper means legislation and the "i" stands for information.

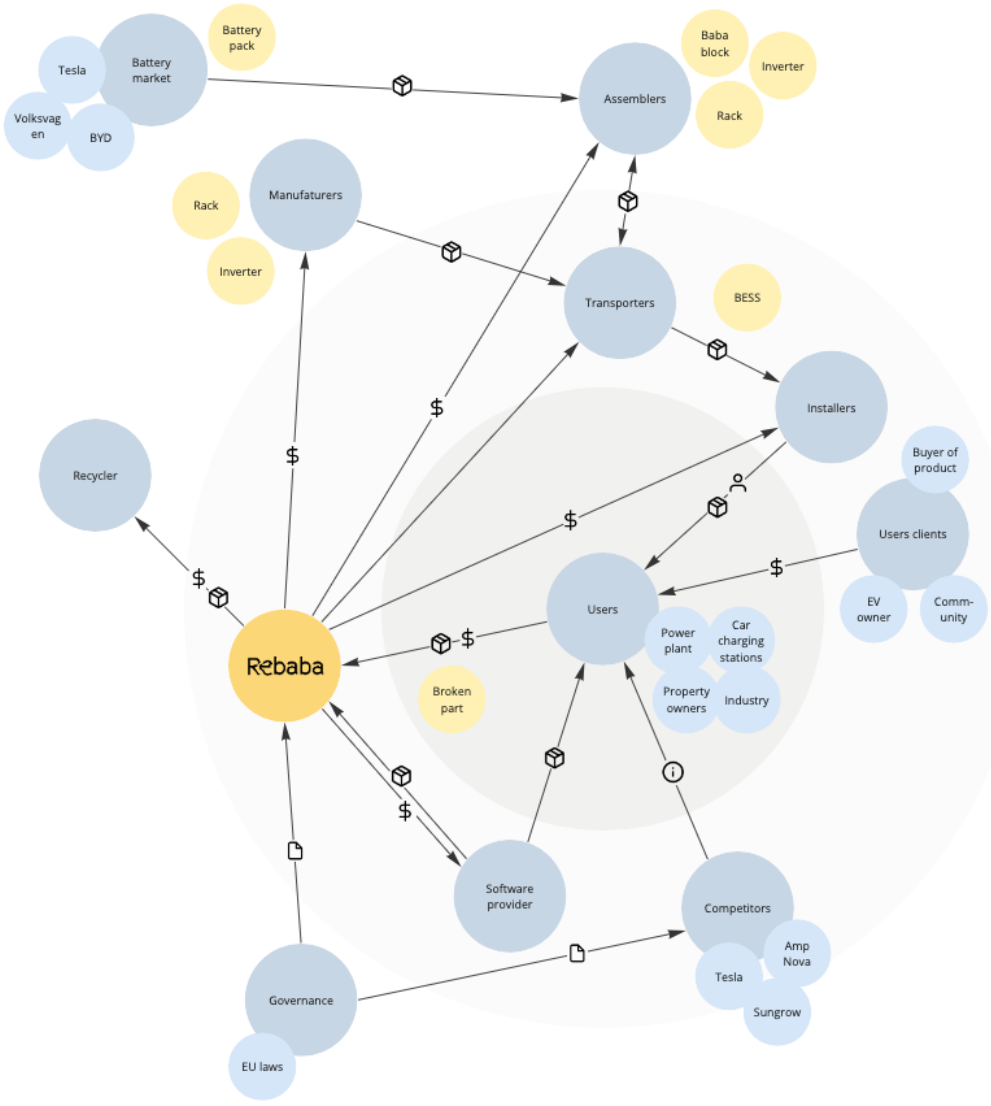
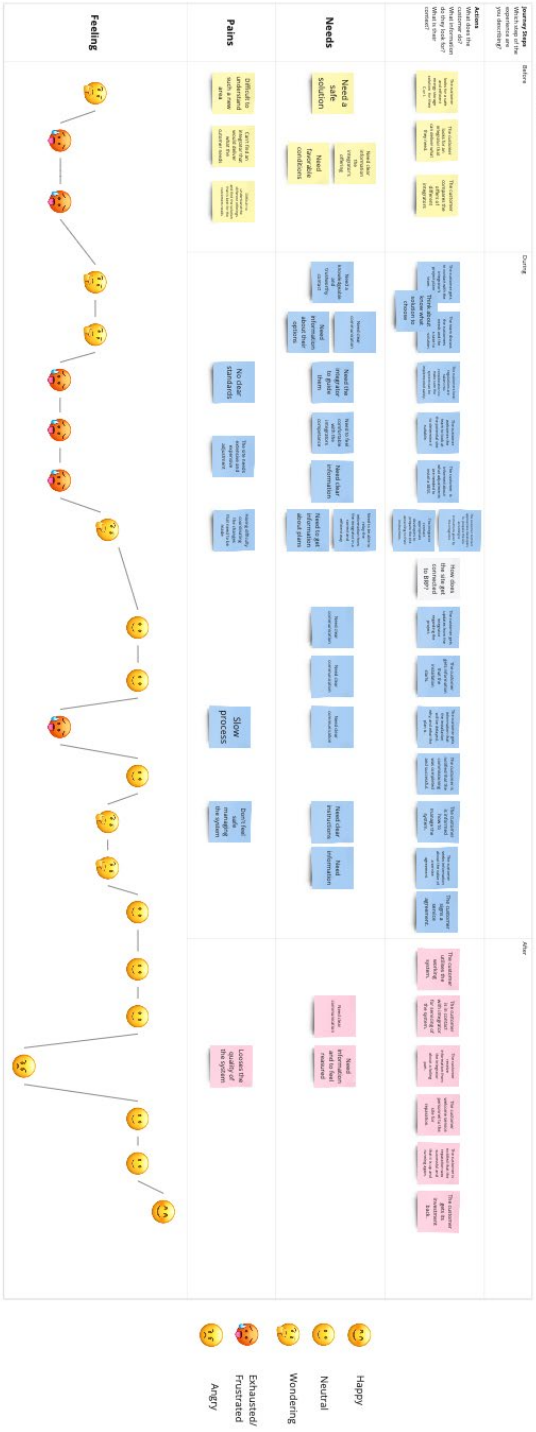


Figure 1. Stakeholder map

Appendix B - Customer Journey



Appendix C - Functional Analysis

Table 1. Functional Analysis of the Battery Energy Storage Solution

Function		Type	Comment	Functional limit
Store	**Electricity**	**MF**		
Provide	Transforming	SF		
Provide	Conversion	SF		
Connect	Batteries	SF		
Provide	Charging	SF		
Provide	Discharging	SF		
Provide	Managing	SF	State of Charge, Energy Flow, Charge and Discharge, Temperature	temp etc.
Provide	Communication	N	With other systems such as the grid operator, to receive commands and send status updates.	
Enable	Remote controlling	W		
Collect	Data	N	Over time to optimise solution	
Provide	Cooling	N		tid temp etc
Provide	Fire extinction	N		tid etc.
Provide	Alerts	N	Before failure, for maintenance and production	
Enable	Repair	N	Through design (modularity?)	
Simplify	Servicing	W	Make vital components easy to access	
Enable	Scalability	W	It should allow for a customisable amount of power storage capacity	
Endure	Weather	N		corr. temp. wetness
Minimise	Environmental impact	W		
Enable	Recycling	N	Material choice and hygiene	
Ergonomics				
Express	Function	N		
Minimise	Cognitive load	W	For users	
Enable	Safe Handling	N	Not endangering user	
Minimise	Physical load	W	For users, using industry standards	
Minimise	Sound	W		..db - db...
Enable	Plug & Play	W	Minimise steps for connecting	
Target Group	**- (C&I)**			
Suit	Target Group	N	Producers and C&I	
Enable	Integration	N	Of other systems, power grid connection (fuses), solar power systems etc.	
Optimise	Ergonomics	N	Interface and intercation with data etc.	
Present	Credibility	W	End user should feel comfortable using the prouct and know that it is working correctly	
Enable	Placing indoors	N	Research shows need for indoors	
Provide	Service	W	BESS solution as a PSS	
Express	Safety	W	Research shows need for clear communication of safety regarding 2nd life batteries	
Market				
Enable	Compatibility	N	Has to be compatible with components such as Invertor from other manufacturer	
Create	Identity	W	The product should be recognisable	
Brand				
Present	Brand Values	N	Sustainable, Disruptive, Genuine Not Compromising, Excluding, Cemented	
Express	Environmental choices	W		
Express	Quality	W		

TRITA – ITM-EX 2024:225
Stockholm, Sweden 2024

www.kth.se